

Auditing Shared Modality-Gap Corrections for Certified Correctness in Frozen CLIP

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Abstract—Randomized smoothing can provide certified ℓ_2 robustness for a smoothed classifier, but a stable prediction need not be a correct one. We study whether post-hoc shared corrections of vision-language prototype geometry improve the certified correctness of frozen zero-shot CLIP classifiers under pixel-space Gaussian smoothing. We first show, through explicit normalized constructions, that a clean global modality-gap summary does not by itself determine the local decision geometry or noisy class probabilities that govern a Cohen certificate. We then derive exact per-draw identities and necessary decision-change conditions for the studied shared prototype operators, while keeping these geometric diagnostics separate from the randomized-smoothing certificate itself. In a preregistered, custody-controlled audit across frozen CLIP backbones and datasets, the registered family of shared/global corrections did not establish an improvement in standard Cohen certified accuracy. In several difficult regimes, raw-clean and smoothed predictions disagree substantially, and certified-wrong predictions concentrate on a small number of classes; we report these observations descriptively rather than as identified causes. We additionally evaluate a separate supervised class-specific positive control to determine whether certified correctness remains recoverable outside the tested shared operator family. We will release registrations, sufficient statistics, and reconstruction code, and argue that evaluations of certified vision-language models should report correctness, stability, abstention, and certified-wrong concentration separately.

Index Terms—randomized smoothing, certified robustness, vision-language models, CLIP, modality gap, preregistration

I. INTRODUCTION

Randomized smoothing turns any base classifier into a smoothed classifier with a certified ℓ_2 radius around each input [1]. For a frozen zero-shot vision-language model such as CLIP [2], the base classifier compares an image feature with a bank of text prototypes, one per class, and the certificate protects whatever class the smoothed classifier selects under Gaussian noise. Throughout, *frozen* means that the pretrained image and text encoders are never updated; the operators we study change only the prototype bank (and, for two-sided centering, the image feature). The certificate therefore says that the selected class is *stable*; it does not say that the selected class is *correct*. Certified accuracy is the fraction of inputs whose stable class is also the true class. Certification of CLIP degrades quickly as the noise grows [3], and in the regime we study many certificates protect a wrong answer.

A natural hope is that the image-text *modality gap* [4], the systematic offset between image and text embeddings, is responsible for part of this loss, and that a cheap post-hoc correction of the prototype geometry could recover certified correctness. The hope has concrete support: Chowers et al.

TABLE I
RESULTS AT A GLANCE: CHANGE IN STANDARD CERTIFIED ACCURACY (POINTS) AGAINST NO CORRECTION, AT RADIUS 0.25 FOR THE AUDIT AND THE PILOT AND AT $r = \sigma$ OTHERWISE. CLASS-SPECIFIC CONTROLS ARE SUPERVISED DIAGNOSTICS, NOT MATCHED COMPARATORS: THEY USE LABELS AND CHANGE CLEAN ACCURACY; THEIR PER-DATASET RANGES SPAN BOTH NOISE LEVELS.

Study (status)	Shared corrections	Class-specific (supervised)
Audit, 12 cells (preregistered)	gate not met; two-sided centering +2.97 (post hoc, failed the clean rule)	—
Pilot, audit regime (exploratory)	supervised shared translation +0.15	+23.45
Extensions A / B, five backbones (registered)	largest of 17 banks +0.32 / +0.75	+15.09 / +13.67
Follow-up A, audit items, fresh noise (registered)	two-sided +2.93 (saved draws +2.97)	—
Follow-up B, fresh items, pooled (registered, primary)	all five primary contrasts within ± 2 ; largest shared estimate +0.50	+5.68 to +9.90
CIFAR-100, not collapsed (secondary)	two-sided +0.29 at $\sigma = 0.12$	+4.67 to +7.71
sealed EuroSAT (secondary)	two-sided +1.87 at $\sigma = 0.25$; descriptively OpenAI backbones +5.50, +4.20, LAION-2B-trained −4.10	+25.20 to +47.87
Imagenette (secondary)	two-sided −1.68 at $\sigma = 0.25$ (near ceiling)	−1.58 to +3.09

prove that translating one modality toward the other improves a nearest-neighbor stability measure under additive embedding noise [5], and mean-centering and modality-wise recentering are already used to improve clean accuracy [6] or empirical robustness [7], [8]. Such corrections need no training and no labels (some use unlabeled images) and apply to a deployed model in one line; if one of them moved a certificate, it would be a cheap deployment-time upgrade of certified trust in a zero-shot classifier. We therefore ask a trustworthiness question about deployed zero-shot classifiers: can a *shared* correction, one common operator applied to every prototype, raise the fraction of inputs whose hard-label Cohen certificate protects the correct class? Table I summarizes the answer.

This paper audits that question rather than proposing a method. A preregistered audit of 18 banks (the identity and 17 shared operators) is the discovery stage; a follow-up registered on fresh items carries the main evidence. The contributions are (i) *mechanism*: when a renormalized common text-side translation can change a hard-label vote, and why the projected-gap translation cannot; (ii) *transportability*: no shared correction gives a consistent gain, and the one post-hoc exception persists

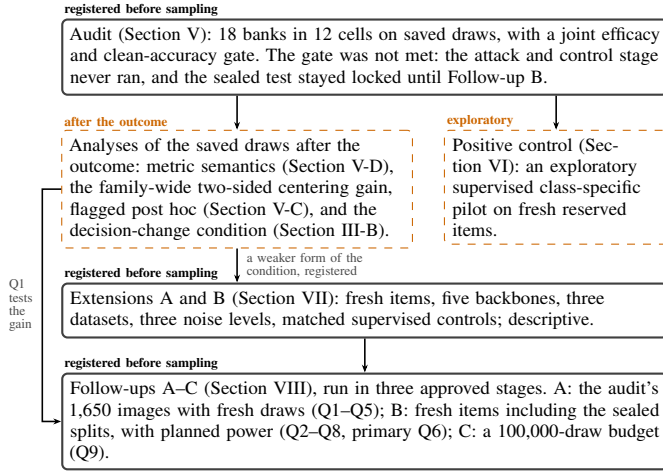


Fig. 1. The studies and what builds on what. Solid boxes were registered, with their analysis, before any outcome was sampled; dashed boxes followed an outcome or are exploratory, and are labeled so wherever used. An arrow means that the later study builds on the earlier one, not the order of execution.

under fresh noise on the audit’s items but changes sign across fresh datasets and checkpoints, while supervised class-specific controls move certified accuracy; and (iii) *evaluation semantics*: standard certified accuracy, clean-anchored stability, abstention and certified-wrong concentration answer different questions. Every registered outcome is reported regardless of sign; Figure 1 shows which studies were registered and what builds on what.

a) Findings: Main result. In the follow-up, registered before its sampling with 7,000 fresh images per noise level, the registered magnitude test places every primary shared contrast within the ± 2 -point margin of no correction at both noise levels under the registered item-pooled mixture of three datasets, and no interval reaches a gain above $+0.68$ points. This is not confined to collapsed cells: on CIFAR-100 at $\sigma = 0.12$, where the smoothed classifier is not collapsed, two-sided centering changes certified accuracy by $+0.29$ points. Supervised class-specific controls, diagnostics rather than matched comparators, gain $+5.68$ to $+9.90$ points (Section VIII). In the descriptive registered extension (five backbones, three datasets, three noise levels) the largest macro increase among the 17 shared banks was $+0.75$ points.

The one exception is setting-dependent. In the preregistered audit the registered family did not establish an improvement under the joint efficacy and clean-accuracy rule. Post hoc, a GR-CLIP-style two-sided centering, the only operator that also recenters the image feature, raised standard certified accuracy at radius 0.25 by $+2.97$ points (simultaneous 95% band $[-0.98, +4.95]$, conditional on the saved draws); it was excluded because one 50-image cell lost 2 images. The follow-up reproduces this gain under fresh noise on the audit’s items; on fresh items it is not uniform: at $\sigma = 0.25$ it recurs, descriptively, on sealed EuroSAT for the audit’s OpenAI backbones and reverses on Imagenette and for the LAION-2B-trained OpenCLIP ViT-B/32.

Mechanism. For a common text-side translation followed

by per-prototype renormalization, Proposition 1 shows that relative scores can change only through class-dependent renormalization, and Proposition 2 gives a necessary condition for a noisy vote to change. For the projected-gap translation of Chowers et al. [5] all renormalization factors are equal, so Corollary 1 gives exact decision invariance under hard-label smoothing. We derived it after the audit; a weaker form, registered before the extension’s sampling, held on fresh noise with 0 exceptions.

Stability is not correctness. Standard and anchored certified accuracy differ by 1.35 to 3.08 points across candidates at radius 0.25, and under the uncorrected model 21.39% of the audit’s positions carry a certificate for a wrong class at radius 0.25, more than are certified correct.

b) Scope: The audit covers OpenAI CLIP ViT-B/32 and ViT-L/14, CIFAR-100 and EuroSAT, three folds, Gaussian noise with $\sigma = 0.25$ in the 224×224 model-input space, and shared prototype corrections under a frozen clean-accuracy eligibility gate. This regime is hard: the uncorrected smoothed classifier is often collapsed, with smoothed accuracy between 12.0% and 43.0% across cells. The extension and the follow-up add a regime where it is not (CIFAR-100 at $\sigma = 0.12$; Imagenette is near ceiling); EuroSAT remains difficult at every noise level. The audit is not an equivalence study, it identifies no causal mechanism, and it says nothing about class-specific, prompt-learned, noise-adapted or image-side methods beyond the positive control.

II. BACKGROUND AND DEFINITIONS

a) Zero-shot prototype classifier: Let ϕ be a frozen CLIP image encoder that maps a model input $x \in \mathbb{R}^{3 \times 224 \times 224}$ to a unit feature $z = \phi(x) \in \mathbb{S}^{d-1}$ (clean inputs lie in $[0, 1]^{3 \times 224 \times 224}$; noisy inputs are not clipped), and let $t_1, \dots, t_K \in \mathbb{S}^{d-1}$ be unit text prototypes obtained by encoding one or more prompts per class and renormalizing their mean. The base classifier is $f(x) = \arg\max_k \langle \phi(x), t_k \rangle$. A prototype operator replaces the bank $\{t_k\}$ by $\{t'_k\}$ and, for two-sided operators, also transforms z ; the base classifier changes accordingly. We write μ_I for the mean image feature over a development split and μ_T for the mean of the K prototypes, and call $\mu_T - \mu_I$ the global modality gap.

b) Adversary and noise coordinates: The adversary is white-box and perturbs the model-input tensor x in ℓ_2 norm; it does not control the Gaussian noise draws, the prompts, or the prototype bank. Noise is added to the 224×224 RGB tensor after deterministic bicubic resizing and center cropping and before channel standardization, and samples are not clipped. All radii in this paper are ℓ_2 distances in that model-input space: a radius of $r = 0.25$ over $3 \times 224 \times 224$ coordinates is a root-mean-square per-coordinate change of 6.4×10^{-4} , below one 8-bit intensity level. Because the CIFAR and EuroSAT images are upsampled from 32×32 and 64×64 pixels, these radii are not native-resolution radii and are not comparable to certificates reported on native CIFAR pixels; Figure 7 (Appendix D) shows one image per dataset at each noise level. A conservative native-resolution bound through the operator norm of the resize map

is possible but we do not claim one, because the resize acts on quantized images.

c) *Cohen certification*: For $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ the smoothed classifier is $g(x) = \operatorname{argmax}_c \Pr[f(x + \varepsilon) = c]$ [1]. The CERTIFY procedure draws n_0 noise samples to select a class $\hat{c}$ (the first maximum of the selection counts, so an exact tie goes to the lowest class index), then draws n independent samples to count votes k for $\hat{c}$ and forms the one-sided Clopper–Pearson lower confidence bound p_L at confidence level $1 - \alpha$ [9]. It abstains if $p_L \leq \frac{1}{2}$ and otherwise certifies radius $R = \sigma \Phi^{-1}(p_L)$: every x' with $\|x' - x\|_2 < R$ satisfies $g(x') = \hat{c}$ with probability at least $1 - \alpha$ over the sampling. This paper uses $n_0 = 128$, $n = 4,096$, $\alpha = 0.001$, and $\sigma = 0.25$ throughout the immutable audit. The confidence statement is pointwise: it holds per input, not simultaneously over a dataset.

d) *Outcome definitions*: Fix a target radius r and a labeled input (x, y) with raw clean prediction $y_0 = f(x)$. Standard Cohen certified accuracy counts the input when the procedure does not abstain, the selected class equals the label, and the radius reaches the target:

$$\text{CA}(r) = \Pr[\text{no abstain, } \hat{c} = y, R \geq r]. \quad (1)$$

The preregistered audit instead governed selection with a stricter operational metric that additionally requires the certified class to equal the candidate’s own raw clean prediction:

$$\text{CA}_{\text{anch}}(r) = \Pr[\text{no abstain, } \hat{c} = y, \hat{c} = y_0, R \geq r]. \quad (2)$$

We call (2) anchored certified accuracy. It is a predicted-label-preservation outcome, not the standard definition, and the two are reported separately everywhere. The audit used the anchored quantity for its registered selection rule; standard Cohen certified accuracy is the primary scientific estimand in the analyses reported here. In (1), (2) and below, $\Pr[\cdot]$ is a rate over the evaluated inputs. We also report smoothed accuracy $\Pr[\hat{c} = y]$ (the selected class is correct, whether or not the procedure abstains), the abstention rate, the raw/smoothed agreement rate $\Pr[\hat{c} = y_0]$, and the certified-wrong rate $\Pr[\text{no abstain, } \hat{c} \neq y, R \geq r]$, the fraction of inputs whose certificate protects an incorrect class.

e) *Aggregation*: A cell is one model–dataset–fold combination. Unless stated otherwise, a macro value is the unweighted mean over the 12 cells, which was the preregistered estimand. Because CIFAR-100 cells contain 500 items and EuroSAT cells 50, we also report item-pooled values where the weighting matters, and we say which weighting a number uses.

III. WHAT A SHARED PROTOTYPE OPERATOR CAN DO TO A NOISY VOTE

A. The registered operator family

All operators are fixed per cell from development data only and then frozen; Figure 2 shows where each acts. With $\text{Normalize}(v) = v/\|v\|$, a coefficient $c \in \{0.25, 0.5, 1\}$ or a step $\eta \in \{0.0025, 0.005, 0.01, 0.02, 0.04\}$, they are:

- *Text mean-centering*: $t'_k = \text{Normalize}(t_k - c(\mu_T - \mu_I))$.
- *Projected-gap translation* (Chowers et al. [5]): $t'_k = \text{Normalize}(t_k + c p)$, where $p = (I - VV^\top)(\mu_I - \mu_T)$ is

the component of $\mu_I - \mu_T$ orthogonal to all differences $t_i - t_j$ (V an orthonormal basis of their span).

- *GR-CLIP-style two-sided centering*: $t'_k = \text{Normalize}(t_k - c\mu_T)$ together with $z' = \text{Normalize}(z - c\mu_I)$ for every clean or noisy image feature; the audit used $c = 1$, and the follow-up also evaluates its text half (prototypes only) and its image half (z only).
- *Clean boundary-active translation*: $t'_k = \text{Normalize}(t_k + \eta u)$ with a unit direction u that increases the normalized clean margin of the development items’ own raw predictions.
- *Noisy-margin translation*: the same form with a unit direction fitted to increase a soft noisy-margin objective over Gaussian draws of the development items at the audit noise level.

Together with the identity this gives the 18 registered banks per cell. Ground-truth labels are never used to build a direction; the two proposal directions target each item’s own raw clean prediction.

B. Per-draw identity and a decision-change condition

Every text-side operator above has the form $t'_k = (t_k + w)/n_k$ with one common translation w and per-prototype renormalization factors $n_k = \|t_k + w\|$. We derived them after the audit outcome was known, so they explain the audit in hindsight rather than predicting it; a weaker form of Proposition 2 was registered as a conformance prediction for the extension of Section VII before its sampling.

Throughout, $z \in \mathbb{S}^{d-1}$ is one image feature (the clean feature or one normalized noisy draw), $s_k = \langle z, t_k \rangle$ are its original scores, $\omega = \langle z, w \rangle$, and every n_k is positive.

Proposition 1 (Per-draw identity). *For every z , $\langle z, t'_k \rangle = (s_k + \omega)/n_k$. Hence for every pair (j, k)*

$$\langle z, t'_j \rangle - \langle z, t'_k \rangle = \frac{n_k(s_j + \omega) - n_j(s_k + \omega)}{n_j n_k}, \quad (3)$$

$$n_j - n_k = \frac{2 \langle t_j - t_k, w \rangle}{n_j + n_k}. \quad (4)$$

Sketch. Expand $\langle z, t_k + w \rangle$, and $n_j^2 - n_k^2 = \|t_j + w\|^2 - \|t_k + w\|^2 = 2 \langle t_j - t_k, w \rangle$ because $\|t_j\| = \|t_k\| = 1$.

Proposition 2 (Necessary condition for a decision change). *Let $s_j \geq s_k$ and suppose that after the translation $\langle z, t'_k \rangle \geq \langle z, t'_j \rangle$. Then*

$$\begin{aligned} 0 \leq s_j - s_k &\leq (n_j - n_k) \langle z, t'_j \rangle \leq |n_j - n_k| \\ &= \frac{2 |\langle t_j - t_k, w \rangle|}{n_j + n_k} \leq \min\{1, \|w\|\} \|t_j - t_k\|. \end{aligned} \quad (5)$$

Consequently, for $\|w\| = \eta$ any pair whose order changes has original cosine margin at most $\eta \|t_j - t_k\| \leq 2\eta$, and if the top-1 label of z changes from j to k , then the original top-1 margin of z (winner minus runner-up) is at most $|n_j - n_k| \leq \max_{i,l} |n_i - n_l|$.

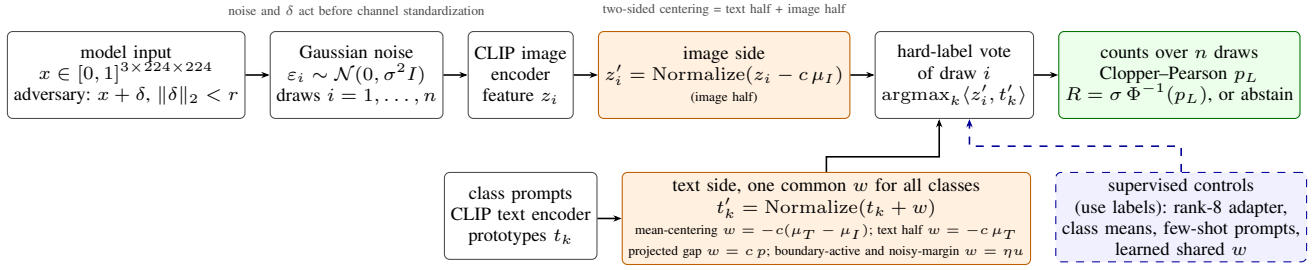


Fig. 2. Where each registered operator acts. The adversary’s perturbation δ and the Gaussian noise live in the 224×224 model-input space. Every shared text-side operator translates all prototypes by one common vector w before renormalization, so by Proposition 2 it can change a draw’s vote only through the renormalization factors; image-side centering changes the feature itself. The supervised controls (dashed) use labels; the class-specific ones bound what is recoverable outside the shared family.

Sketch. Rearranging the hypothesis with (3)–(4) gives the first two inequalities, $|\langle z, t'_j \rangle| \leq 1$ gives the third, and the last one is Ptolemy’s inequality for the points $t_j, t_k, -w, 0$ together with the reverse triangle inequality (Appendix B). The second inequality also carries a direction: when $\langle z, t'_j \rangle > 0$ and $s_j > s_k$, the original winner can lose only to a class whose translated raw norm is strictly smaller, $n_k < n_j$.

Corollary 1 (Projected-gap translation is decision-invariant). *If $\langle w, t_i - t_j \rangle = 0$ for all i, j , as for the projected-gap translation $w = c \cdot p$ at every coefficient c , then all n_k equal one value n and $\langle z, t'_k \rangle = (s_k + \omega)/n$ with n and ω independent of k , so the argmax (including ties) is unchanged for every z . In exact arithmetic with a fixed tie rule, on any fixed set of noisy draws the operator therefore reproduces the uncorrected hard labels, vote counts, selected classes, certified radii, and accuracies exactly: it is an exact invariance control, not a competing method.¹*

Sketch. $n_k^2 = 1 + 2\langle t_k, w \rangle + \|w\|^2$ and $\langle t_k, w \rangle = \langle t_1, w \rangle$ for every k .

Corollary 2 (Finite flip budget). *Fix N draws $z_1, \dots, z_N$, let v_i be the number of draws whose original argmax is i , let $k_0 = \max_i v_i$, and let E_F be the number of draws whose original winner j has a competitor $k \neq j$ with $s_j - s_k \leq |n_j - n_k|$; E_F is at most the number of draws whose original top-1 margin is at most $\max_{i,l} |n_i - n_l|$. After the translation the counts v'_i satisfy $v'_i \leq v_i + E_F$ for every i , hence $\max_i v'_i \leq k_0 + E_F$. In particular, if $k_0 + E_F < k_{\min}$ for a confirmation threshold $k_{\min}$, then no class reaches $k_{\min}$ on these draws after the translation.*

Sketch. By Proposition 2 every flipped draw is one of the E_F eligible draws, and a class gains at most one vote per flipped draw. Because E_F depends only on the original scores and on the normalizers n_k recorded when a bank is built, the corollary can rule out a threshold crossing on these draws, without rescoring, whenever $k_0 + E_F < k_{\min}$; when the inequality fails, it does not show that a crossing occurs. Saved top-1 margins

give a looser upper bound on E_F , and the exact pairwise count needs the original score differences. The certificate itself is still computed only from the hard-label counts by the separate Cohen procedure; the bound is not a certificate.

Remark 1 (Two-sided operators). *The GR-CLIP-style operator also maps $z \mapsto z' = \text{Normalize}(z - \mu_I)$; we write the audit’s coefficient $c = 1$, and a general c replaces μ_I and μ_T by $c\mu_I$ and $c\mu_T$. Renormalizing z never changes an argmax, but the shift does. Assume $z \neq \mu_I$ and $t_k \neq \mu_T$ for every class k , so that both normalizations are defined. With $n_k = \|t_k - \mu_T\| > 0$,*

$$\begin{aligned} \langle z', t'_k \rangle &= \frac{\langle z - \mu_I, t_k - \mu_T \rangle}{\|z - \mu_I\| n_k} \\ &\propto \frac{s_k - \langle z, \mu_T \rangle - \langle \mu_I, t_k \rangle + \langle \mu_I, \mu_T \rangle}{n_k}, \end{aligned} \quad (6)$$

where the proportionality constant is common to all classes. The class-dependent term $-\langle \mu_I, t_k \rangle$ is a per-class bias that no common text translation produces, so Propositions 1–2 do not bound this operator: it can change the label of a draw whose original margin exceeds every bound in (5). This is why, within the audit’s registered family, it is the only operator that can act where the text-only translations cannot.

Remark 2 (Engine boundary). *Nothing in this section is a certificate; certified radii are computed separately from hard-label counts as in Section II.*

Two consequences matter for reading the results. A step of $\eta = 0.0025$ can flip a noisy draw only if its original top-1 margin is at most $|n_j - n_k| \leq 2\eta$. The bound defines which draws are eligible; how many are eligible depends on the empirical margin distribution, and on the saved draws the registered steps changed very few certified outcomes (Section V-B). The projected-gap translation leaves every hard label unchanged for every coefficient, so under hard-label smoothing it is an exact negative control rather than a competing method; this is a property of the projection Chowers et al. designed to preserve nearest neighbors, made explicit for the smoothed classifier. Within the audit’s registered family, only the two-sided operator, which also shifts the image feature, escapes the bound.

¹This is the mechanism behind the identical aggregates of the three projected-gap coefficients and of the uncorrected bank in the saved audit record; in floating-point arithmetic only numerical ties could differ.

C. Why a clean global-gap summary cannot settle the question

One might hope to predict a certificate from the clean geometry alone. It cannot be done in general. Appendix A gives an explicit smooth normalized encoder and two binary prototype banks that share the encoder, the clean feature of the certified input, the complete global-gap vector under any common image-reference distribution, the clean margin and the prototype correlation, yet have population Cohen radii that differ by an arbitrary factor. The two banks are mirror images of one another, and the only asymmetry is how the encoder’s response direction meets each decision boundary; the listed clean summaries do not contain that response. This is a scoping remark, not a theorem about CLIP: it motivates measuring noisy votes rather than clean gaps.

IV. PREREGISTERED AUDIT DESIGN

a) Models, data and roles: The audit, registered in a development stage and a validation stage whose results were then frozen, uses OpenAI CLIP ViT-B/32 and ViT-L/14 with frozen weights and frozen prompts: the single OpenAI CIFAR-100 prompt and a three-template EuroSAT ensemble [2], [10], [11]. From the CIFAR-100 training split and the EuroSAT image folder we drew three folds, each with five development and five validation images per class, giving 500 validation items per CIFAR-100 cell and 50 per EuroSAT cell, 12 cells and 3,300 validation positions in total. Development items served only to build operators, and validation items to select and evaluate them. The CIFAR-100 test split and a EuroSAT holdout were sealed and were not read by the audit, the positive control or the extension, whose CIFAR-10 items also come from the training split; the registered follow-up of Section VIII later spent the sealed EuroSAT holdout (1,000 items) and 3,000 CIFAR-100 official-test items once, as its registration specifies.

b) Certification: Every candidate bank in every cell was certified on every validation item with independent selection and confirmation noise streams derived from separate base seeds, so all 18 banks of a cell see identical draws (a paired design), while selection draws are never reused for confirmation. In total the audit evaluated 59,400 candidate–item positions with 7,603,200 selection votes and 243,302,400 confirmation votes. Radii are reported at $r \in \{0, 0.1, 0.25, 0.5\}$ with $r = 0.25$ preregistered as primary.

c) Selection and efficacy rule: The protocol selected one proposal objective and step by the macro *anchored* certified accuracy at $r = 0.25$ over the 12 cells, with frozen tie-breaks, and selected one coefficient per tunable baseline by the same metric. A candidate was eligible only if its macro clean-accuracy drop relative to the uncorrected model was at most one point and no single cell dropped by more than two points. The efficacy gate then required the selected proposal to exceed the eligible mean-centering comparator on the macro metric and on every model and dataset marginal. If the gate failed, the protocol stopped: the preregistered attack and control stage and the sealed test remained locked, and the project committed to reporting the negative in full. That stage never ran; the

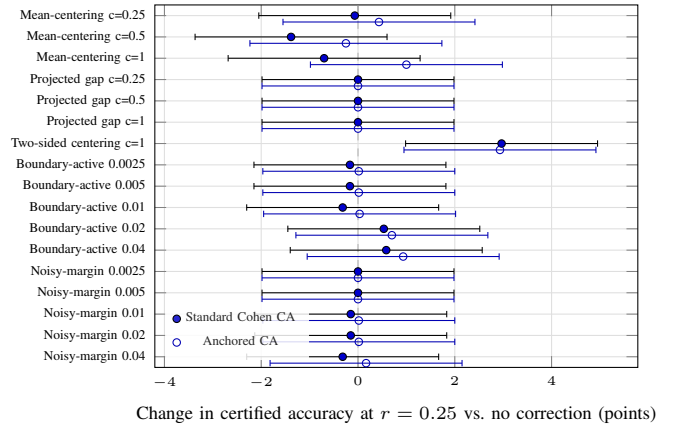


Fig. 3. Change in certified accuracy against the identity (no correction) for each of the 17 other registered banks, at the primary radius, under both metrics. Bars are the common 95% simultaneous max-absolute-deviation band over all 136 contrasts. At this radius only the two-sided centering operator’s band excludes zero; the projected-gap banks are exactly zero; the proposal banks change at most 18 of the 3,300 certified outcomes.

sealed test was later allocated, by a registration hashed before sampling, to the follow-up (Section VIII).

d) Custody: Configurations, split assignments, seeds, prompts and every bank were hashed before sampling; the reconstruction checks are in Appendix E.

e) Post-hoc audits of the immutable result: Three analyses were specified after the outcome was known and are labeled as such. (i) *Metric semantics:* standard certified accuracy (1) was reconstructed beside the anchored metric (2) for all candidates, cells and radii. (ii) *Complete-family bands:* a paired item bootstrap, stratified by dataset, fold and class, resampled the same items across both models, all candidates, radii and metrics, and produced one unstudentized 95% simultaneous max-absolute-deviation band over all 136 candidate–radius–metric contrasts against the uncorrected model (100,000 replicates, 330 strata, seed fixed before execution). The band is a conditional compatibility statement about the saved sample; it is not an equivalence test. (iii) *Exact-count adjudication:* the clean gate was recomputed with integer counts because the implementation compared floating-point percentages.

V. AUDIT RESULTS

A. The preregistered gate failed

Figure 3 and Table IV (Appendix C) report every registered bank at the primary radius under both metrics with its post-hoc simultaneous band. The saved selection was the clean boundary-active bank at step 0.0025, with anchored certified accuracy 8.43% against 8.42% for no correction, and standard certified accuracy 11.30% against 11.47%. No mean-centering coefficient passed the clean gate, so the protocol’s fallback comparator was the projected-gap bank at coefficient 0.5, which is identical to no correction in every decision in exact arithmetic (Corollary 1); the saved proposal exceeded it by one position in anchored certified accuracy, the protocol’s selection metric, while its standard certified accuracy was lower. The

efficacy rule also required strictly positive margins against mean-centering on every model and dataset marginal; computed for the exact-count proposal against each of the three mean-centering coefficients, all 3 of these forced contrasts failed on at least one marginal, one only by a single position. The negative decision is therefore a failure to establish efficacy under the frozen joint rule. It is not evidence that the effect is zero.

a) *Exact-count adjudication*: The implementation compared percentage differences in floating point, so a one-image drop in a 50-image cell was represented as 2.0000000000000018 rather than 2.0 and was rejected. Re-computing the written rule with integer counts makes 8 further banks eligible and changes the selected proposal to step 0.04 (standard 12.05%, anchored 9.35%). The gate decision does not change under either reading; the saved selection is preserved and both readings are reported.

B. The proposal banks were near-identity operators

Proposition 2 gives a necessary condition for vote changes, which the follow-up finds loose (Section VIII); the proposals were also small. The saved proposal moved the prototype bank by a macro Frobenius displacement of 0.016, against 7.011 for text mean-centering at coefficient one and 6.680 for two-sided centering; the largest step reaches 0.259. On the saved draws, the six proposal banks with steps up to 0.01 change the standard certified outcome at $r = 0.25$ of at most 3 of the 3,300 positions, the boundary-active step-0.04 bank changes 18, and the two-sided operator changes 192. The three projected-gap banks give the same certified outcomes and reported aggregates as the uncorrected model, as Corollary 1 predicts in exact arithmetic; a few saved confirmation counts differ slightly, consistent with floating-point numerical ties (on fresh draws the follow-up bounds every such change by a registered tie budget, Q4 in Section VIII), and their largest absolute contrast is 0.00 points. The registered family thus contained 15 distinct classifiers, of which the text-only proposals changed very few saved certified outcomes and the largest text-only translations damaged clean accuracy.

The common band, a single-step resampling band in the sense of Westfall and Young [12], has a half-width of 1.98 points set by the noisiest contrasts (the large-displacement comparators at small radii), many times the paired standard error of the proposal contrasts; it is uninformative about the proposals, and the counts above describe them.

C. One shared operator did move certified correctness

In this post-hoc complete-family analysis, whose bands are simultaneous and conditional on the saved draws, the GR-CLIP-style two-sided centering is, at the primary radius, the only bank whose simultaneous interval excludes zero under either metric: $+2.97$ points of standard certified accuracy at $r = 0.25$ with band $[+0.98, +4.95]$, and $+2.93$ points anchored with band $[+0.95, +4.92]$. Across the whole family, 8 contrasts exclude zero: 6 of the eight contrasts of this operator, all positive, and 2 negative standard-metric contrasts of text mean-centering

at $r = 0$ (Section V-D). It is therefore the only bank with a positive interval that excludes zero. Its macro clean accuracy rose to 64.92% from 60.88%, and clean accuracy improved in 9 of 12 cells. It was ineligible because one cell (ViT-L/14, EuroSAT, fold 1) lost 4.00 points, that is 2 of 50 images.

Three qualifications bound the size of this effect without removing it. The gain is $+0.60$ points on the six CIFAR-100 cells, where 2 of six cells move down, and $+5.33$ points on the six 50-image EuroSAT cells; the band that excludes zero is carried by the EuroSAT cells (Appendix C). Second, every bootstrap stratum contains five items, and resampling five items with replacement understates the variance by a factor of $4/5$; inflating the band accordingly gives $[+0.75, +5.18]$, which still excludes zero. Third, the preregistered estimand weights the 12 cells equally, so the EuroSAT cells carry half the estimand (Kish design effect 3.025, effective sample size 1,090.9); the item-pooled gain is $+1.03$ points. The operator is also the only one in the audit’s registered family that transforms the image feature, which is exactly the case that Proposition 2 does not bound. We do not conclude that two-sided centering is effective in general: within the audit, the registered clean gate rather than the certificate excluded it, and the follow-up finds it in its original setting, while on fresh items it lies within the ± 2 -point margin under the pooled estimand and changes sign across datasets and checkpoints (Section VIII).

D. Stability is not correctness

For every candidate, at $r = 0.25$ anchored certified accuracy lies 1.35 to 3.08 points below standard certified accuracy, because the anchored metric additionally discards positions whose stable class is correct but differs from the raw clean prediction: for the uncorrected model, 66 of the 499 certified-correct positions at $r = 0.25$ are such anchor mismatches. The two metrics can also disagree. Text mean-centering at coefficient one has a standard contrast of -3.30 points at $r = 0$ with band $[-5.28, -1.32]$, which excludes zero, while its anchored contrast at the same radius is -0.78 points with band $[-2.77, +1.20]$; it raises raw/smoothed agreement to 39.27% from 21.92% while lowering clean accuracy to 47.33%. An evaluation that reports only one of the two metrics can therefore reach a different conclusion about the same operator.

Table II partitions the 3,300 positions of the uncorrected model at $r = 0.25$ (item-pooled). Abstentions account for 821 positions; 1,677 nonabstaining positions select a class other than the label, and 706 of them carry a certificate at this radius; 303 select the correct class but do not reach the radius; and 499 are certified correct. Under the anchored metric, 1,688 nonabstaining positions are anchor mismatches and 433 remain anchored and certified correct. Wrong smoothed classes are the dominant loss, and the certified-wrong rate, 21.39% of all positions, exceeds the certified-correct rate. Within a cell, the single most frequent class among certified-wrong positions accounts for 9.7% to 100.0% of them. Across cells, raw clean accuracy ranges from 42.0% to 75.4%, smoothed accuracy from 12.0% to 43.0%, raw/smoothed agreement from 2.0% to 40.4%, the share of positions with zero radius

TABLE II
FAILURE PARTITION AT RADIUS 0.25 POOLED OVER 3,300 POSITIONS FOR NO CORRECTION AND THE GR-CLIP-STYLE TWO-SIDED CANDIDATE, UNDER THE STANDARD METRIC AND THE ANCHORED METRIC. COUNTS COME FROM THE AUDIT’S SAVED PER-EXAMPLE SUFFICIENT STATISTICS AND REPRODUCE ITS PER-CELL FAILURE CATEGORIES.

Outcome at $r = 0.25$	No corr. n	%	GR-CLIP n	%
<i>Standard metric</i>				
Abstained	821	24.88	815	24.70
Nonabstaining, wrong smoothed class	1,677	50.82	1,681	50.94
of which certified wrong at 0.25	706	21.39	719	21.79
Correct smoothed class, radius < 0.25	303	9.18	271	8.21
Certified correct at 0.25	499	15.12	533	16.15
<i>Anchored metric</i>				
Abstained	821	24.88	815	24.70
Nonabstaining anchor mismatch	1,688	51.15	1,635	49.55
Anchor preserved, raw class wrong	103	3.12	112	3.39
Anchor correct, radius < 0.25	255	7.73	246	7.45
Anchored certified correct at 0.25	433	13.12	492	14.91

under the anchored metric from 62.8% to 98.0%, and the largest single smoothed-class share from 5.4% to 84.0%. These are descriptions of where certificates are lost in this regime, consistent with the high-noise degradation reported by [3]; the controlled experiments that would identify a cause were never run, and we make no causal claim.

a) *Finite confirmation budget*: Every estimate here concerns the finite-budget procedure. With 4,096 confirmation draws, fewer than the 10^5 of Cohen et al. [1], the exact bound censors radii: at $r = 0.25$, 17 uncorrected positions and 19 for the saved proposal. Pairing does not equalize this censoring; Q9 measures a larger confirmation budget directly (Section VIII).

VI. POSITIVE CONTROL: IS CERTIFIED CORRECTNESS RECOVERABLE AT ALL?

A negative result for shared operators says little unless some intervention can move the certificate in the same regime. We therefore report a separate, exploratory, supervised 20-shot class-specific pilot, originally developed as a candidate class-specific method and used here only as a positive control, on fresh reserved items of the same two backbones, datasets and noise level, with 1,000 CIFAR-100 and 100 EuroSAT confirmation items per cell and the same certification budget. Using 20 labeled images per class, it trained class-specific low-rank tangent adjustments of the prototype bank in the spirit of parameter-efficient adaptation [13] under a noisy-and-clean cross-entropy objective. Seven variants were evaluated (Table III): a supervised *shared* translation (one common correction, the comparator); a *low-rank* adapter with the cross-entropy objective only; a *tail-only* variant that adds a loss on the vote fraction near the certification threshold; the *full* objective at rank 8, at rank 4, and in a direct (non-tangent) parametrization; and a *pseudo-label* variant of the full objective trained on the model’s own clean predictions instead of labels. The pilot was registered with its own configuration and custody package, but its planned smoke run was skipped, so we call it exploratory.

Table III shows that class-specific adaptation recovers a large share of certified correctness in a regime where shared

translations did not: standard certified accuracy at $r = 0.25$ rises from 11.60% to 35.05% for the full rank-8 variant, a paired macro gain of +23.45 points with 95% interval $[-20.25, +26.63]$, while the supervised shared translation reaches only 11.75%. The certified-wrong rate falls from 34.18% to 10.43%. Four qualifications bound the reading. The tail-only ablation reaches 36.50%, above the full variant’s point estimate, with a certified-wrong rate of 9.05%, and the full variant’s contrast against it is -1.45 points with interval $[-4.05, +1.13]$; that interval does not establish a difference, so the additional certificate-aware terms are not shown to be necessary. Clean accuracy also rises, from 60.88% to 68.58% (and to 70.80% for the tail-only variant), so these results do not isolate a benefit specific to certificate targeting from ordinary accuracy improvement under supervised adaptation. The pseudo-label variant, the only one that uses no ground truth, reaches 27.10% standard certified accuracy (+15.50 points, $[-13.27, +17.73]$) but lowers clean accuracy to 46.98%, a drop of 13.90 points that fails the clean gate: it buys certified stability at the cost of clean correctness, which is the trade this paper argues must be reported explicitly. Finally, the pilot itself was not compared with noise-aware prompt learning [14], the natural baseline for any text-side adaptation; the follow-up adds a few-shot prompt control inspired by it (Section VIII). The control therefore establishes a scope boundary: the certificate in this regime is movable by class-specific supervised adaptation, and the result of Section V concerns shared operators only.

VII. REGISTERED EXTENSION: FIVE BACKBONES, THREE DATASETS, THREE NOISE LEVELS

After the audit, we registered a separate extension with hashed configurations (Table VI, Appendix C) before sampling any of its evaluation outcomes, to probe the scope of the earlier findings; every registered cell completed and is reported regardless of sign. It re-estimates the same 18 banks on a fresh development split, adds four prospectively trained supervised controls with 20 labels per class (a learned shared translation in two forms, a noisy class-mean bank, and a rank-8 class-specific tangent adapter), and evaluates all 22 banks on fresh items in two registered experiments: Extension A, the two audit backbones at $\sigma \in \{0.12, 0.5\}$ on CIFAR-100, EuroSAT and CIFAR-10 (12 primary cells plus two CIFAR-10 bridge cells at $\sigma = 0.25$), and Extension B, OpenAI ViT-B/16, OpenAI RN50 and an OpenCLIP ViT-B/32 trained on LAION-2B [15], [16] at $\sigma \in \{0.25, 0.5\}$ (18 cells), with 500 items per CIFAR cell and 150 per EuroSAT cell and the same 1,150 unique items reused across cells. The registered primary outcome per experiment is the equal-cell macro change in standard certified accuracy at $r = \sigma$ against no correction, with one shared-item class-stratified simultaneous band over the 21 contrasts.

a) *Regime*: Table VII (Appendix C) reports the uncorrected model in every cell. At $\sigma = 0.12$ the CIFAR cells are far from collapsed: smoothed accuracy ranges from 40.00% to 90.00% and standard certified accuracy at $r = \sigma$ from 26.00% to 81.80%, while the EuroSAT cells remain difficult (9.33% to 26.67% smoothed). At $\sigma = 0.5$ every dataset collapses. The

TABLE III

POSITIVE-CONTROL PILOT: MACRO STANDARD CA, ANCHORED CA AND CERTIFIED-WRONG RATE AT RADIUS 0.25 AND MACRO CLEAN ACCURACY OVER 4 CELLS (1,000 CIFAR-100 AND 100 EUROSAT CONFIRMATION ITEMS PER CELL, 20 SUPERVISED SHOTS PER CLASS), WITH ALL FIVE PLANNED PAIRED CONTRASTS AND THEIR CLASS-STRATIFIED PERCENTILE BOOTSTRAP 95% INTERVALS (10,000 REPLICATES). CLEAN GATE: MACRO DROP AT MOST 1 POINT AND EVERY CELL DROP AT MOST 2 POINTS.

Candidate	Std CA	Anch. CA	Clean	Cert.-wrong	Clean gate
No correction	11.60	8.43	60.88	34.18	pass
Shared translation (noisy CE)	11.75	8.68	61.28	33.43	pass
Low-rank r=8 (noisy CE)	27.10	23.35	72.50	13.58	pass
Tail-only r=8	36.50	28.75	70.80	9.05	pass
Full objective r=4	33.35	24.60	62.15	11.78	pass
Full objective r=8	35.05	29.55	68.58	10.43	pass
Full objective, direct	34.60	31.20	66.23	11.43	fail
Full objective r=8, pseudo-labels (no ground truth)	27.10	21.15	46.98	17.75	fail

Planned contrast (Std CA)	Δ	95% lower	95% upper
Full objective r=8 vs. no correction	+23.45	+20.25	+26.63
Full objective r=8 vs. tail-only r=8	-1.45	-4.05	+1.13
Full objective r=8 vs. full objective r=4	+1.70	-0.65	+4.00
Full objective r=8 vs. low-rank r=8 (noisy CE)	+7.95	+4.73	+11.15
Low-rank r=8 (noisy CE) vs. shared translation (noisy CE)	+15.35	+12.70	+18.03

extension therefore contains the non-degenerate regime that the immutable audit lacked.

b) Outcomes: Table VIII (Appendix C) lists every registered contrast. In the audit backbones (Extension A) the largest increase among the 17 banks of the shared grid is +0.32 points (noisy-margin step 0.04 and two-sided centering $c=1$) and the smallest change is -2.56 points (mean-centering (text) $c=1$, with clean accuracy -12.31 points); at a common half-width of 1.84 points, 1 of the 17 grid bands excludes zero (mean-centering (text) $c=1$). Two-sided centering changes standard certified accuracy by +0.32 points ($[-1.52, +2.16]$) and clean accuracy by +4.66 points. In the three further backbones (Extension B) the largest increase in the shared grid is +0.75 points (mean-centering (text) $c=0.5$), the smallest -0.15 (boundary-active step 0.02), 0 of 17 grid bands exclude zero with half-width 1.56 points, and two-sided centering changes standard certified accuracy by +0.42 points ($[-1.14, +1.98]$). Against the registered ± 0.5 -point reporting region (prediction X1; the registrations number their predictions X1 and X3 to X5), 15 and 15 of the 17 grid changes lie inside it (Table VIII). In the two CIFAR-10 bridge cells at the audit's noise level two-sided centering changes certified accuracy by +2.40 points (ViT-B/32) and -0.80 points (ViT-L/14).

The supervised controls behave differently. The learned shared translation, which like the shared family applies one common vector to every prototype (its tangent form rescales each prototype by a factor fixed by that vector; Appendix B) but is fitted with labels, changes certified accuracy by -0.02 points ($[-1.86, +1.82]$) in Extension A and +0.96 points in Extension B, so prediction X3 (the shared control stays inside the band), scored as registered on the tangent form, was confirmed in both Extension A and Extension B; the pure form, which reuses its vector, changes it by -3.26 ($[-5.10, -1.42]$) and -0.08 points ($[-1.64, +1.48]$). The class-specific rank-8 adapter changes certified accuracy by +15.09 points ($[+13.26, +16.93]$, clean +10.22 points) and +13.67 points ($[+12.11, +15.24]$), and the noisy class-mean bank, which replaces text prototypes by

image-side class means, by +9.12 points while changing clean accuracy by -39.44 points; prediction X4 (at least one per-class control leaves the band upward) was confirmed in both Extension A and Extension B, and both per-class controls do so in both experiments. Prediction X5 (the identity bank's top-class share and predicted-class Gini rise with σ in every model and dataset) was supported in Extension A but not uniformly supported in Extension B: in Extension B the share fell between $\sigma = 0.25$ and $\sigma = 0.5$ in several RN50 and ViT-B/16 cells; we report the decline without an explanation.

c) Reading: On fresh items, across five backbones and three noise levels, including cells where the smoothed classifier is not collapsed, the largest macro increase in the 17-bank shared grid was +0.32 points (audit backbones) and +0.75 points (further backbones), and the separately trained shared tangent control changed the macro by -0.02 and +0.96 points, with bands containing zero in both experiments. The largest grid decrease in Extension A, -2.56 points, came from the largest text-side translation, which also lost clean accuracy; in Extension B that translation changed certified accuracy by +0.61 points and clean accuracy by -7.67 points. Class-specific capacity fitted with the same supervision moved certified accuracy by +15.09 and +13.67 points. The extension is descriptive: its cell sizes of 500 and 150 items were fixed without a registered power calculation, the power of any contrast depends on the paired discordance and on the dependence between repeated items, its bands are conditional on the realized draws, and no equivalence is claimed. Execution deviations and verification status are in Appendix E.

d) Fresh-noise mechanism diagnostic: The four-cell diagnostic at $\sigma = 0.25$ records per-draw features so that the flip condition can be checked directly. Its registered conformance prediction (every draw whose label changes under a shared translation satisfies the inequality of Proposition 2) recorded 0 exceptions; the descriptive flip-budget partition and the useful/harmful taxonomy are in Table IX (Appendix C).

VIII. PROSPECTIVE FOLLOW-UP: FRESH NOISE, FRESH ITEMS, LARGER BUDGET

After the extension we registered a three-part follow-up (Table X) before sampling any of its outcomes, to answer the questions the audit left open. Follow-up A re-certifies the audit’s 1,650 images with the audit’s saved bank tensors, fresh certification draws and a fresh pipeline; it is a same-item re-certification, not an independent replication. Follow-up B uses fresh, never-used items: 3 backbones (the two audit backbones and an OpenCLIP ViT-B/32 trained on LAION-2B [15], [16]), CIFAR-100 test (3,000 items), sealed EuroSAT (1,000) and Imagenette, a ten-class ImageNet subset [17] (3,000), at $\sigma \in \{0.12, 0.25\}$. Its registered planning calculation (Table X; not the operating power of the bootstrap) gives the pooled family at each σ (7,000 unique images) a minimum detectable effect of 1.3 points and the EuroSAT family alone 3.47. Its primary estimand is item-pooled, whereas the audit’s was equal-cell; both are reported. Follow-up C re-certifies 100 items per dataset with 100,000 confirmation draws to test the 4,096-draw budget. Every cell of Follow-ups A and B evaluates the same 36 banks (Follow-up C, three of them): the identity; text mean-centering, the projected-gap translation, two-sided centering and its text-only and image-only halves at 3 coefficients each; both proposal objectives at 8 steps up to 0.32; and 4 supervised controls. Inference uses a class-stratified shared-item bootstrap with 100,000 replicates, Holm-adjusted direction tests, and practical-magnitude classes against a smallest effect of interest of 2 points, fixed with the planning calculation before sampling; the margin is a registered convention, not a claim that smaller changes are unimportant.

The registered questions are: whether the post-hoc two-sided gain persists under fresh draws (Q1); how much of it the text and image halves carry (Q2); whether text-side steps change only draws that Proposition 2 allows, and whether the projected-gap bank is decision-invariant draw by draw (Q3, Q4); what the supervised controls achieve (Q5); whether any shared operator changes standard certified accuracy on fresh items, and by at least 2 points (Q6, primary); whether the result persists where the smoothed classifier is not collapsed (Q7); whether the radii are comparable to the literature (Q8); and how many certificates change with the larger budget (Q9). Every question is reported below or in Appendix D, whatever its outcome. The registered analysis ran once, unchanged, after a recorded analysis-stage approval; cells not run: 0 of 12 in Follow-up A and 0 of 18 in Follow-up B (Appendix E).

a) Audit items under fresh draws (Follow-up A): Under fresh draws, two-sided centering changes the audit items’ equal-cell macro certified accuracy by +2.93 points ([+1.52, +4.35]), against +2.97 points on the saved draws; the registered Q1 verdict is *reproduced*: a positive direction ($p < 0.001$) and no detected shortfall ($p = 0.484$), which is not a claim of equality. Item-pooled, the change is +0.97 points ([+0.22, +1.75]) against +1.03 (reproduced). In the item-pooled decomposition (Q2) the text half alone changes certified accuracy by −0.27 points, the image half by +1.55, and their interaction is −0.30.

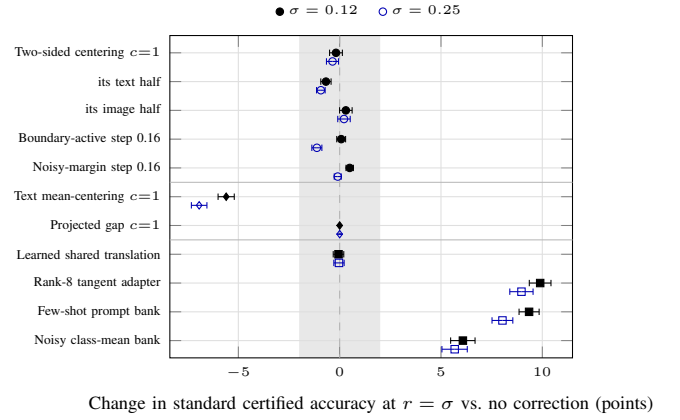


Fig. 4. Registered Follow-up B contrasts on fresh items, pooled over datasets and backbones: change in standard certified accuracy at $r = \sigma$ against no correction, with unadjusted 95% bootstrap intervals, at $\sigma = 0.12$ (filled) and $\sigma = 0.25$ (open). The shaded band is the registered smallest effect of interest, ± 2 points. Circles are the registered primary family (magnitude classes in Table XI); diamonds are two unsupervised reference banks; squares are the supervised controls, which use labels. The rows were fixed before any outcome existed.

b) Fresh items (Follow-up B, primary): Figure 4 shows the registered primary family (Q6): two-sided centering at $c = 1$, its text half, its image half, and the two proposals at step 0.16, each against no correction; Table XI (Appendix D) gives both estimands and the adjusted classes. Two-sided centering changes standard certified accuracy at $r = \sigma$ by −0.18 points ([-0.50, +0.13]) at $\sigma = 0.12$ and by −0.35 ([-0.65, −0.05]) at $\sigma = 0.25$ (equal-cell −0.24 and +0.14); its Holm-adjusted classes are unresolved in direction and practically equivalent in magnitude at both noise levels. At $\sigma = 0.12$ the Holm-adjusted direction is positive for the noisy-margin proposal at step 0.16 and negative for the text half of two-sided centering at $c = 1$; at $\sigma = 0.25$ it is negative for the text half of two-sided centering at $c = 1$ and the boundary-active proposal at step 0.16. Direction and magnitude answer different questions: a contrast can be resolved in sign and still lie inside the margin. Descriptively, the unadjusted 95% intervals of all ten primary pooled contrasts lie within ± 1.37 points, and none reaches a gain above +0.68 points. Across the 5 primary contrasts the magnitude classes are 0 gain, 0 loss, 5 practically equivalent and 0 inconclusive at $\sigma = 0.12$, and 0, 0, 5 and 0 at $\sigma = 0.25$. Among all unsupervised banks the largest point estimates are +0.50 (the noisy-margin proposal at step 0.16) and +0.45 points (the image half of two-sided centering at $c = 0.5$), and the smallest −5.60 (text mean-centering at $c = 1$) and −6.93 (text mean-centering at $c = 1$). The text half alone changes certified accuracy by −0.68 and −0.93 points at the two noise levels, the image half by +0.30 and +0.21, and their interaction is +0.19 and +0.36 (Q2; every coefficient in Table XIII). The supervised controls (Q5), the regime of the uncorrected model (Q7), the orientation against published radii (Q8) and, descriptively, the certified-accuracy curves of four banks fixed in advance (Figure 5) are in Appendix D.

c) *Draw-level checks and budget*: Q3 (text-side steps change only draws that Proposition 2 allows) conforms in both Follow-ups, with no exception. Its coarse form is necessary but loose: on the audit’s items the coarse flip budget admits 44.8% of draws at step 0.0025 and 99.9% at step 0.04, whereas 0.21% to 5.50% change, and useful and harmful changes nearly cancel (85,677 against 81,651 for the boundary-active proposal at step 0.04; Table XIV). Q4 (the projected-gap bank is decision-invariant draw by draw) conforms in both Follow-ups, with 235 (A) and 1,150 (B) changed draws, all within the registered numerical tie budget: no correction’s margin between the two classes involved was at most 1.03×10^{-5} and 1.03×10^{-5} (the tolerance plus the largest renormalization difference), a bound that does not identify the cause of a change. With 100,000 instead of 4,096 confirmation draws (Q9, Follow-up C), the uncorrected model’s certified-correct count at $r = \sigma$ over its 6 cells moves from 208 to 211 of 600 (3 gained, 0 lost); over the 3 registered banks 5 of 1,800 certificates change, at most 3 for no correction. The two registered determinism checks, exact identity with Follow-up B of the confirmation prefix and of the full certificate inputs, returned *yes* and *yes*.

d) *Reading*: Under fresh draws the post-hoc two-sided gain on the audit’s items persists (+2.93 against +2.97 points), matching its registered prediction, so it is not specific to the saved certification draws; it is carried by the image half and, descriptively, by the EuroSAT cells (+5.33 against +0.53 points on CIFAR-100). On fresh items it does not transport as a uniform gain. Under the registered item-pooled mixture of the three datasets (weights 3:1:3) it lies within the registered ± 2 -point margin at both noise levels and, in an unregistered comparison, below the fresh-draw estimates on the audit’s items at $\sigma = 0.25$ (item-pooled -0.35 against $+0.97$ points, equal-cell $+0.14$ against $+2.93$); the sign of this small aggregate even depends on the weighting. Its dataset and backbone estimates range from positive to negative: on sealed EuroSAT at that σ its direction is positive ($+1.87$ points; descriptively $+5.50$ and $+4.20$ for the two audit backbones but -4.10 for the LAION-2B-trained model), and on Imagenette it is negative (-1.68). All five registered primary contrasts are classified as practically equivalent to no correction at both noise levels, with small resolved effects of both signs, and no unsupervised bank’s pooled estimate exceeds $+0.50$ points; the only contrast classified as a dataset-level gain under its registered within-family procedure, $+3.73$ points for the noisy-margin proposal at step 0.16 on the sealed EuroSAT holdout at $\sigma = 0.12$, comes with a clean-accuracy change of -10.47 points there, and the clean rule flags it. Where the classifier is not collapsed and supervision moves the certificate, on CIFAR-100 at $\sigma = 0.12$ (smoothed accuracy 30.80% to 62.57%), two-sided centering changes certified accuracy by $+0.29$ points; Imagenette is near ceiling (at least 81.43% certified correct at $r = \sigma$), and even the supervised controls move it by only -1.58 to $+3.09$ points. The controls that change the prototypes class by class gain $+5.68$ to $+9.90$ points pooled, mostly on EuroSAT and with clean-accuracy changes of their own (Appendix D), whereas one label-fitted shared vector changes certified accuracy by

-0.06 and -0.03 under the item-pooled estimand but by $+3.13$ and $+2.53$ on EuroSAT. The follow-up therefore delimits the audit’s post-hoc exception rather than overturning it: the gain persists in its original setting but has no stable, transportable direction on fresh items, and the pooled result places it within the registered margin without implying a zero effect in every dataset or checkpoint.

IX. RELATED WORK

a) *Randomized smoothing and certified accuracy*: Randomized smoothing certifies an ℓ_2 radius for the Gaussian-smoothed classifier, which returns the class most probable under input noise [1], [18], [19]. Cohen et al. give the radius $\sigma \Phi^{-1}(p_L)$ from a lower confidence bound p_L and the CERTIFY procedure, which selects a class from one set of draws and bounds its probability from a second [1]. Salman et al. train the base classifier against attacks on the smoothed model [20]; Yang et al. characterize the radii attainable under other norms and noise shapes [21]; Kumar et al. certify average class scores and margins [22], so the use of confidence or margin information inside smoothing is established. In this literature, certified accuracy counts an input only when the smoothed class equals the label and its radius reaches the target; it does not ask whether the smoothed class agrees with the noise-free prediction of the base model. We keep this estimand and report the raw-clean-anchored variant as a separate operational quantity. That smoothing shrinks decision regions unevenly across classes is known [23], and abstention-aware certified metrics exist for segmentation [24]; the certified-wrong rate we report is a reporting recommendation grounded in these observations, not a new metric.

b) *Certified vision-language classification*: Nirala et al. certify open-vocabulary CLIP classifiers with randomized smoothing, reuse earlier computation when prompts change, and report that certified accuracy degrades sharply as the noise grows [3]. Our audit runs in a high-noise regime of the kind they describe, and we do not present that degradation as a finding. Seferis et al. certify generative vision-language models by mapping their outputs to an oracle classification task [25]; we study frozen prototype classifiers. PromptSmooth learns textual prompts [26] under Gaussian noise so that a frozen medical vision-language model certifies better [14]; it is the direct precedent for any text-side adaptation of a smoothed CLIP classifier, and the shared operators we audit are far more constrained than learned prompts. Carlini et al. denoise the input with a diffusion model before a pretrained classifier [27]; this image-side route lies outside our operator family and is not evaluated. In embedding space, Pautov et al. smooth normalized embeddings and convert an embedding-boundary distance into an ℓ_2 radius for prototype classifiers [28], and RetrievalGuard certifies smoothed nearest-neighbor retrieval [29]. Xia et al., in a preprint under review, bound cosine feature deviation by a Gaussian robustness score and convert it into a top-one prototype certificate through the clean margin and prototype correlation [30]. None of these feature certificates enters our estimates; every certificate we report is a hard-label Cohen

certificate of the smoothed version of its frozen classifier (the supervised controls are not zero-shot).

c) *The modality gap and its corrections*: Liang et al. define the modality gap of contrastive encoders such as CLIP [2], attribute it to initialization and contrastive optimization, and show that shifting it changes zero-shot behavior in a task-dependent way [4]. Schrodi et al. find that a few dimensions drive the gap and that its influence on accuracy is often overshadowed by other factors [31], and Fahim et al. show that contrastive training separates even samples of one modality [32]; Levi and Gilboa show that CLIP’s image and text embeddings lie on linearly separable ellipsoid shells not centered at the origin [33]. That the global gap magnitude does not explain accuracy is therefore established. Post-hoc corrections form the operator families we audit. Text-side mean-centering and the two-sided variant that also recenters image features are normalized zero-shot adaptations of the GR-CLIP calibration [6]. Chowers et al. treat the gap as a robustness feature: under a centered-orthogonality assumption and additive noise in embedding space, translating one modality toward the other provably improves a nearest-neighbor stability measure, and a projected translation preserves clean nearest neighbors [5]. Our audit does not contradict that theorem. It concerns a different object, the hard-label vote of a classifier smoothed with pixel noise, and shows that the normalized zero-shot adaptation of their projected operator cannot change any such vote. Sato et al. show that clean gap reduction can change class-relative decisions and induce prediction hubness [34]; the certified-wrong class concentration we report arises without any correction and under Gaussian pixel noise, and must not be read as correction-induced clean hubness. Empirical robustness work aligns image and text representations during robust fine-tuning or by training-free projection [7], [8], [35], [36] or fine-tunes the vision encoder adversarially [37]. These methods target attack accuracy rather than Cohen certificates and lie outside our frozen shared operator family.

d) *Preregistration and post-selection inference*: Nosek et al. argue that preregistration separates hypothesis generation from hypothesis testing [38]. Kuchibhotla et al. review the bias that arises when the same data select a candidate and estimate its effect, and the simultaneous-inference remedies [39]. Our validation split both selected and evaluated candidates, so we report the complete family under one simultaneous band and describe the selected candidate rather than test it. Lakens notes that an equivalence claim needs a prespecified smallest effect of interest and a dedicated procedure [40]; the audit registered neither and claims no absence of effect, and the follow-up registered both before its sampling (Section VIII).

e) *What is and is not new*: New here are a necessary decision-change condition for a renormalized common text-side translation under hard-label smoothing, stated per draw, with the corollary that the projected-gap operator is decision-invariant; a metric-semantics audit that keeps standard certified accuracy, raw-clean-anchored stability, abstention, and certified-wrong concentration separate and shows that these estimands rank operators differently; and a preregistered audit that reports the

complete operator family with simultaneous bands, including the one operator whose band excludes zero at the primary radius, followed by a follow-up, registered with a planning power calculation before its sampling, that delimits that operator’s gain on fresh items. Not new, and not claimed, are certified zero-shot or open-vocabulary VLM classifiers and their degradation at high noise [3]; mean-centering and projected gap corrections [5], [6]; noise-aware prompt adaptation [14]; finite-sample certification machinery [1]; class-specific low-rank adaptation [13], used here only as a supervised positive control; and feature-space or embedding certificates [28], [30].

X. LIMITATIONS AND LESSONS FOR CERTIFIED-VLM EVALUATION

A. Limitations

The extension and the follow-up address the audit’s main gaps (a non-collapsed regime, fresh items, a planned sample size); the limitations that remain are these.

- *Noise regime*. In the audit, Gaussian noise with $\sigma = 0.25$ is added in the model-input space after bicubic upsampling of small native images. Our radii are distances in that model-input space; because the resize acts on quantized images, no exact conversion to native-resolution radii exists, and our radii are not comparable with certificates reported at native resolution. Smoothed accuracy collapses in this regime, as Nirala et al. observed for CLIP at larger noise [3]: without correction it ranges across cells from 12.0% to 43.0%, and raw/smoothed agreement from 2.0% to 40.4%.
- *Breadth*. The audit covers two OpenAI CLIP backbones, two datasets, frozen prompt templates, and one noise level in 12 cells. The registered extension and follow-up add backbones, datasets, and noise levels; the extension’s cell sizes were fixed without a registered power calculation, and its cells reuse the same 1,150 items, so its contrasts are dependent; it is reported descriptively and regardless of sign. In the follow-up, the non-collapsed evidence rests on CIFAR-100 at $\sigma = 0.12$: Imagenette is near ceiling, where even the supervised controls barely move the certificate, and the dataset families are secondary. The follow-up spent the sealed EuroSAT holdout, so a later confirmatory study needs new items. Nothing here transfers to other prompts, class-conditioned or prompt-level methods, image-side denoising, native-resolution regimes, or certificates other than hard-label Cohen certificates, such as soft-score or feature-space ones.
- *Power*. The audit’s band (half-width 1.98 points) is uninformative about the proposals, which moved standard certified accuracy by -0.17 and $+0.58$ points; under-powered paired comparisons are common in this field [41]. The band conditions on the realized votes and establishes neither equivalence nor a uniform minimum detectable effect; a finite-stratum correction widens it to 2.22 points. Equal-cell weighting has Kish design effect 3.025 (effective sample size 1,090.9 of 3,300 positions),

so item-pooled aggregates are reported alongside. The follow-up registered a planning power calculation before sampling; it assumes a paired discordance and is not the operating power of its bootstrap analysis.

- *Gate design.* The per-cell clean rule’s two-point tolerance is one image in a 50-image cell, and it decided the two-sided operator’s exclusion. The production gate compared in floating point and rejected exact two-point drops for 8 candidates (exact-count re-adjudication is reported alongside), and the positive decision path required an eligible mean-centering comparator, none of which existed, so the forced fallback comparison was decided at single-image resolution.
- *Post-selection.* The validation split both selected candidates and produced the reported outcomes. We therefore report all 18 registered candidates under one simultaneous band, and describe rather than test the selected candidate. The audit registered no smallest effect of interest, so it makes no equivalence or evidence-of-absence claim [40]; the follow-up registered one (2 points) before sampling, and only its registered magnitude classes use it.
- *Confirmation budget.* Each position used 4,096 confirmation draws at $\alpha = 0.001$. At radius $r = \sigma$, 17 no-correction positions have a plug-in estimate that would certify the radius but a finite-sample lower bound that does not; at radius zero, 111 positions select the correct class but abstain. In Follow-up C, 100,000 instead of 4,096 draws changed 5 of 1,800 certificates at $r = \sigma$; the selection budget was not varied.
- *Positive control.* The class-specific low-rank control is supervised with 20 labeled images per class. Its clean accuracy rises from 60.88% to 68.58%, so its certified gain is not isolated from ordinary clean-accuracy improvement under supervision, and the full-versus-tail-only interval does not establish a difference (−1.45 points, interval −4.05 to +1.13). It bounds what remains recoverable outside the shared family; it does not rescue that family and is not proposed as a method. In the follow-up, a few-shot prompt control inspired by noise-aware prompt learning [14] gains about as much as its rank-8 adapter.
- *Causal identification.* Active, inactive, random, and sign-reversed control interventions were never run. Anchor disagreement, certified-wrong concentration, candidate displacement, and vote collapse are descriptive observations, not identified causes.
- *Certificate confidence.* Each certificate holds with probability at least $1 - \alpha$ for its own position. The fraction of certified positions is not a family-wise event, and no simultaneous coverage across positions is claimed.
- *Verification.* All verification was internal to the author team: other authors re-evaluated or replayed every result, several checks with AI assistance, labeled as such in the review ledger; no party outside the team has reproduced the chain (Appendix E).

B. Lessons

The audit suggests a minimal reporting template for certified vision-language classifiers.

- 1) Report standard certified accuracy in the sense of Cohen et al. [1], and label any raw-clean-anchored or predicted-label-stability variant as a separate quantity.
- 2) Report abstention and the certified-wrong rate at every radius, together with the class concentration of certified-wrong predictions.
- 3) Report raw/smoothed agreement per cell.
- 4) Report both equal-cell and item-pooled aggregates, with the design effect of the chosen weighting, and per-dataset and per-backbone effects beside pooled ones.
- 5) State eligibility rules as exact counts, state the per-cell granularity in images, and never compare thresholds in floating point.
- 6) Report the complete registered family with simultaneous bands, including inert and ineligible members.
- 7) State the noise coordinate system and the native resolution, and say whether reported radii are comparable across resolutions.

XI. CONCLUSION

We audited whether shared corrections of CLIP’s prototype geometry improve the certified correctness of a frozen zero-shot classifier under randomized smoothing. A per-draw identity shows that a renormalized common text-side translation can change a hard-label vote only through the difference between prototype renormalization factors, which makes the projected-gap translation exactly inert under hard-label smoothing. The preregistered audit did not establish an improvement under its joint rule; its one post-hoc exception, two-sided centering, persisted under fresh noise on the audit’s items. A follow-up registered before its sampling delimits that exception: on fresh items every primary shared contrast lies within the ± 2 -point margin under the registered item-pooled estimand, while at $\sigma = 0.25$ the gain recurs, descriptively, on sealed EuroSAT for the audit’s backbones and reverses on Imagenette and for the LAION-2B-trained model. Supervised class-specific controls raise certified accuracy, with clean-accuracy changes of their own; a label-fitted shared vector does not. The evidence marks a transportability boundary rather than an impossibility: shared corrections can move certified correctness in particular settings, but not as a consistent gain across datasets and checkpoints. A certificate supports trust only when an evaluation also reports whether the stabilized prediction is correct and how the conclusion varies across data and models (Section X-B).

OPEN SCIENCE

a) *Release boundary:* The anonymous artifact linked from the submission form is assembled from hash-identified files by the artifact deadline, and its manifest lists every file with its SHA-256. The table below states, for each group of claims, the inputs the artifact provides, what those inputs let a reviewer recompute on a CPU, and what is omitted. A hash identifies an omitted file; it does not make that file replayable.

Claims	Released inputs; recomputable on a CPU	Omitted, listed by hash
Audit metrics, bands and gate (Sec. V)	Protocol and registrations; per-example selection and confirmation counts, certified radii and clean predictions for all 18 candidates in all 12 cells; candidate metadata, bank hashes and development tensors; the simultaneous-analysis configuration and output. Recomputes every audit table and macro, the complete-family band and the exact-count gate.	None
Theory (Sec. III-B, App. A–B)	Decision-change note, verification scripts and tests, non-identification constructions. Recomputes every numerical check of the propositions.	None
Positive control (Sec. VI)	Configuration snapshot, split indices, trained adapters, confirmation outputs, analysis report and custody manifest. Recomputes every aggregate and the paired bootstrap.	Feature cache (about 0.8 GB)
Extension (Sec. VII)	Registrations and code; per-item vote counts of all 36 cells with receipts and recovery records; prototype banks; registered analysis outputs; reconstruction package. Recomputes every extension aggregate and band.	Development and control-training caches (about 5.8 GB)
Fresh-noise diagnostic (Table IX)	Registration, code and the recorded diagnostic outputs. Regenerates the table from those outputs only.	Per-draw features (about 3.0 GB); independent recomputation needs them
Follow-up (Sec. VIII)	Registrations, the three recorded stage approvals (approver names redacted in the anonymous version), code, data preflight and bank manifest; per-item vote counts of every cell with their custody manifests; registered analysis outputs. Recomputes every registered output from the vote counts.	Prepared prototype banks and preparation caches, if they exceed the repository limit; recomputing them needs the public checkpoints and datasets

b) What else is not released, and why: The omitted archives exceed the anonymous repository’s size limits and will be deposited in an archival repository with the final version. Model weights and datasets are public third-party assets: the OpenAI CLIP checkpoints [2] and the LAION-2B-trained one [15] are loaded through OpenCLIP [16], and the CIFAR-100, EuroSAT, CIFAR-10 and Imagenette [17] images are obtained from their public distributions. The artifact references them by hash rather than redistributing them. The sealed EuroSAT holdout and the CIFAR-100 official test split were not read by the audit, the positive control or the extension. The registered follow-up spent the sealed EuroSAT holdout in full and 3,000 CIFAR-100 official-test items, each once and for evaluation only, and their item lists are released with that experiment; no later experiment may use them for selection.

c) Preregistration: The preregistration underlying this audit is voluntary and self-enforced through hashed registrations and fail-closed launch guards; it was checked within the author team, not by a venue. The design follows the pre-registration

format used at machine-learning workshops [42]: hypotheses, analysis rules and stop rules were fixed and hashed before sampling, and every registered outcome is reported regardless of its direction.

d) Layout: The artifact is organized as `registrations/` (protocol, pre-analysis and extension registrations with hashes), `results/exp017/` (per-cell sufficient statistics, candidate metadata and bank hashes), `analysis/` (the metric-semantics audit, the simultaneous-band output and configuration, the exact-count adjudication), `paper/` (source, generators, generated tables and figure data, tests, lint), `theory/` (decision-change note and verification tests), `control/` (the positive-control release), `extension/` (code, registrations, per-item vote counts, prototype banks, registered analysis outputs and the reconstruction package), `followup/` (registrations, stage approvals, per-item vote counts with their custody manifests, and the registered analysis outputs), and `review/` (the review ledger: for every check, its reviewer, the reviewed version, its scope, its date, and whether it was AI-assisted; the anonymous version names reviewers by role). Files that could identify the authors (project trackers, logs, hostnames) are excluded.

e) Study names: The artifact’s README maps each study named in the paper (the audit, the positive control, Extensions A and B, the fresh-noise diagnostic, and Follow-ups A to C) to its files.

f) Reproduction without a GPU: The generators rebuild every table, macro and figure from the released analysis outputs and record the hash of each source. The audit’s aggregates and bands are recomputed from its per-example sufficient statistics, the positive control’s from its confirmation outputs, and the extension’s from its vote counts; independent recomputation of the fresh-noise diagnostic needs the separately archived per-draw features. None of these steps downloads a model, opens a sealed split, or draws a new scientific sample. A GPU is needed only to regenerate the features and votes themselves; the pinned environment, seeds, and hashes for that step are recorded, and regenerated prototype banks can be checked against the released hashes. Regenerating the audit’s per-example tables needs CPU PyTorch to read the saved statistics; every other step needs only NumPy and SciPy. The follow-up’s outputs are recomputed from its vote counts. The released bundles carry no version-control history, so their hashes identify the released bytes but do not by themselves attest when or by whom those bytes were produced. Each registered cell was sampled under its registration, and no run was repeated or extended after outcomes were inspected; the recoveries and the recomputation described in Appendix E preceded any inspection of outcomes.

LLM USAGE CONSIDERATIONS

LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality.

Beyond editing, large language model assistants were used during this project, through chat interfaces (Claude and ChatGPT) and as coding agents (Claude Code and Codex). They

drafted code, derivations, documentation and review packaging, edited text, and performed adversarial technical reviews of the code, the review packages and drafts of this paper. The authors therefore state the boundary of that use. The authors determined the research questions, the experimental design, the acceptance criteria and the final theoretical claims, made every decision, and checked the assistants’ outputs, including the drafted code and derivations. No reported numerical result is an LLM-generated value: all reported values are produced by deterministic scripts from recorded experimental data, and each generated value records the hash of its source. Model-drafted analysis code was tested before use and is released with the artifact. No large language model was used to generate, select, or filter experimental data, and no such model is part of the certification procedure. The review ledger released with the artifact lists, for every check reported in this paper, who performed it (by role in the anonymous version), on which immutable version, with what scope and on which date, and marks each AI-assisted review as such. The human authors are responsible for all submitted content and results.

a) Motivation: We chose these assistants for their strength in coding, writing and technical checking, and used them for the tasks listed above: code, drafts of derivations and documentation, review packaging, text editing and adversarial technical review.

b) Limitations: Hosted-model outputs are not deterministic and cannot be replayed; the models’ sizes, weights and serving hardware are not public; and their outputs can be wrong in ways that read as plausible. For these reasons no model output is evidence in this paper: every reported value comes from deterministic scripts over recorded data, model-drafted code is tested before use, and AI-assisted reviews are labeled as such and are not counted as human verification.

c) Compute and environmental footprint: No language model was trained or fine-tuned for this work, and none ran on the authors’ own hardware; all use went through the providers’ hosted services. The providers do not disclose the hardware or the energy behind a query, so we cannot report either. Keeping the research questions, the design and the decisions with the authors limited the queries to tasks whose outputs the authors could check.

The experiments themselves used no language model. Their compute is reported in Appendix C: the audit took 14.4 GPU-hours on one RTX 4090, the extension was planned at 23 GPU-hours on the same GPU, and the follow-up sampled for 15.0 hours on seven RTX 5090 GPUs and 15.3 hours on one more. This compute follows from the certification procedure, in which every certificate needs 128 selection and 4,096 confirmation encodings of noisy inputs, and from the follow-up’s planning calculation, which fixed its sample size. We limited it by encoding each noisy draw once and scoring every bank of a follow-up cell on that encoding, by restricting the 10^5 -draw budget study to 6 cells of 100 items and 3 banks, and by releasing vote counts and sufficient statistics, so that every table is recomputed on a CPU without new sampling.

ETHICAL CONSIDERATIONS

This work uses only publicly distributed third-party datasets (CIFAR-10, CIFAR-100, EuroSAT, and Imagenette, a ten-class ImageNet subset) and public model checkpoints, involves no human subjects, and develops no offensive capability. Its findings concern the evaluation of certified classifiers: a certificate that protects a wrong class can create unwarranted trust, and the reporting template in Section X-B is intended to reduce that risk. The negative result is reported in full, including the one registered operator with a positive simultaneous band that excludes zero at the primary radius, so that the paper does not understate what shared corrections can do.

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APPENDIX A

CLEAN-GAP NON-IDENTIFICATION CONSTRUCTIONS

This appendix records the Cohen construction of Section III-C and a companion example for a feature-space certificate. Both are scoping remarks: they show that clean summaries cannot determine a certificate, not that any correction fails.

A. Binary Cohen construction

Proposition 3 (Clean-gap non-identification for Cohen smoothing). *Let $0 < \gamma < 1$, $s = \sqrt{1 - \gamma^2}$, $b, d > 0$, and $\kappa > 0$ with $\kappa \neq 1$. Put*

$$z_0 = \frac{(b, b, d)}{\sqrt{2b^2 + d^2}}, \quad q = \frac{b}{\sqrt{2b^2 + d^2}},$$

and use the shared scalar-input encoder

$$f(u) = \text{Normalize}(z_0 + u(e_1 + \kappa e_2)).$$

Consider the two binary unit-prototype banks

$$t_{1,\pm} = \gamma e_3 \pm s e_1, \quad t_{2,\pm} = \gamma e_3 \pm s e_2.$$

Let P_X be any common fixed image-reference distribution for which $\mu_I = \mathbb{E}_{X \sim P_X}[f(X)]$ exists, and define the global gap as the prototype mean minus μ_I . The banks have the same prototype mean and hence the same complete global-gap vector. At $u = 0$ they also have the same positive clean pairwise margin and within-bank prototype correlation. Under $u = \epsilon \sim \mathcal{N}(0, \sigma^2)$, however, their population Cohen randomized-smoothing radii are $R_1 = q$ and $R_2 = q/\kappa$. Thus these shared clean summaries do not identify the population Cohen radius.

Proof. All prototypes have norm one and both bank means equal γe_3 . Because the encoder and P_X are common, μ_I is common, so both global gaps equal $\gamma e_3 - \mu_I$. Their clean positive margins at $u = 0$ are both $2sq > 0$, and both prototype correlations equal $\gamma^2 - s^2$. Since normalization has a positive common denominator, bank one predicts the positive class exactly when $q + u > 0$, whereas bank two does so exactly when $q + \kappa u > 0$. Hence their positive-class probabilities are $p_1 = \Phi(q/\sigma)$ and $p_2 = \Phi(q/(\kappa\sigma))$. In the binary case the population Cohen radius is $\frac{\sigma}{2}[\Phi^{-1}(p) - \Phi^{-1}(1 - p)] = \sigma\Phi^{-1}(p)$, giving the stated radii. The unnormalized encoder vector has constant third coordinate $d/\sqrt{2b^2 + d^2} > 0$, so the normalization is globally defined and the encoder is smooth and globally Lipschitz. $\square$

a) Reading the construction honestly: The reflection that swaps e_1 and e_2 fixes z_0 and maps bank one onto bank two, so the clean query and the two banks are reflection-related: the prototype geometry and the listed clean summaries at $u = 0$ coincide, and the global-gap vectors coincide because the two systems share the image mean μ_I (one encoder) and both bank means equal γe_3 ; the reflection itself need not fix μ_I . This does not make arbitrary clean statistics of the two systems equal, because the shared encoder is not reflected. It responds to the input along $e_1 + \kappa e_2$, and at $u = 0$ the derivatives of the two banks’ pairwise margins differ by $2s(1 - \kappa) \neq 0$. This local response, relative to each decision boundary, is the quantity the listed summaries omit. Moreover $R_1 = q$ for every σ : in this one-dimensional setting the smoothed classifier equals the base threshold classifier and the radius is the distance to the threshold. The proposition therefore says precisely that the listed clean feature-space summaries do not contain the encoder’s local response, which is all the audit needs it to say.

b) Class-balanced reference set: For a finite empirical convention, take $u_+ = 0$ and $u_- = -L$ with $L > \max\{q, q/\kappa\}$, label them positive and negative, and weight the two classes uniformly. Both banks classify both reference inputs correctly and share $\mu_I = [f(0) + f(-L)]/2$; hence their class-balanced global-gap vectors are identical while the certified query at $u = 0$ retains the unequal radii. The released tests verify the probabilities, radii, Lipschitz bound and reference set numerically, including an encoder-level Monte Carlo check.

c) Finite-sample distinction: The proposition concerns population class probabilities. A finite-sample Cohen certificate replaces the unknown selected-class probability by an exact one-sided confidence lower bound and is therefore conservative and random.

B. Feature-space certificate example

The feature-space smoothing certificate of [30], an under-review preprint, certifies a prototype prediction from a Gaussian robustness score of the image feature and a geometric threshold of the prototype bank. A scalar-input step encoder with two prototype banks that are rotations of one another holds the complete global gap and the exact population robustness score fixed while the geometric threshold, and hence the certified lower bound, varies with the rotation angle. In that example the smoothed decision itself is the same for every angle, so the true robust radius does not change; only the certificate’s slack does. We record the example as certificate-bound variation and do not use it to make any claim about robustness. Neither construction transfers any feature-space threshold or decomposition to the Cohen certificate, and the smoothed-embedding certificates of [28] are a third, separate engine.

APPENDIX B PROOFS

Throughout, $t_1, \dots, t_K \in \mathbb{S}^{d-1}$, $w \in \mathbb{R}^d$, $n_k = \|t_k + w\| > 0$, $t'_k = (t_k + w)/n_k$, $z \in \mathbb{S}^{d-1}$, $s_k = \langle z, t_k \rangle$ and $\omega = \langle z, w \rangle$; all norms are ℓ_2 norms.

Proof of Proposition 1. By linearity, $\langle z, t'_k \rangle = \langle z, t_k + w \rangle / n_k = (s_k + \omega) / n_k$. Subtracting the expressions for j and k over the common denominator $n_j n_k$ gives (3). For (4), $n_k^2 = \|t_k\|^2 + 2\langle t_k, w \rangle + \|w\|^2 = 1 + 2\langle t_k, w \rangle + \|w\|^2$, so $n_j^2 - n_k^2 = 2\langle t_j - t_k, w \rangle$; dividing by $n_j + n_k > 0$ gives $n_j - n_k = 2\langle t_j - t_k, w \rangle / (n_j + n_k)$. $\square$

Proof of Proposition 2. Step 1 (rearrangement). By Proposition 1 the hypothesis reads $(s_k + \omega) / n_k \geq (s_j + \omega) / n_j$. Multiplying by $n_k > 0$ gives $s_k + \omega \geq (n_k / n_j)(s_j + \omega)$, and subtracting $s_j + \omega$ from both sides,

$$\begin{aligned} s_k - s_j &\geq \left(\frac{n_k}{n_j} - 1 \right) (s_j + \omega) \\ &= (n_k - n_j) \frac{s_j + \omega}{n_j} = (n_k - n_j) \langle z, t'_j \rangle, \end{aligned}$$

that is, $s_j - s_k \leq (n_j - n_k) \langle z, t'_j \rangle$. Multiplying the hypothesis by n_j instead of n_k gives the companion bound $s_j - s_k \leq (n_j - n_k) \langle z, t'_k \rangle$. The lower bound $0 \leq s_j - s_k$ is the assumption $s_j \geq s_k$.

Step 2 (Cauchy-Schwarz). Since z and t'_j are unit vectors, $|\langle z, t'_j \rangle| \leq 1$, hence $(n_j - n_k) \langle z, t'_j \rangle \leq |n_j - n_k|$.

Step 3 (exact form). By (4), $|n_j - n_k| = 2|\langle t_j - t_k, w \rangle| / (n_j + n_k)$.

Step 4 (reverse triangle inequality). $|n_j - n_k| = \left| \|t_j + w\| - \|t_k + w\| \right| \leq \|t_j - t_k\|$.

Step 5 (Ptolemy's inequality). We claim $|n_j - n_k| \leq \|w\| \|t_j - t_k\|$. This is trivial for $w = 0$. For $w \neq 0$ use the inversion $x \mapsto x^* = x / \|x\|^2$ of nonzero vectors, which satisfies $\|x^* - y^*\| = \|x - y\| / (\|x\| \|y\|)$ (expand both squared norms). Apply the triangle inequality to the images of the three nonzero points t_j , t_k and $-w$; since $t_j^* = t_j$ and $t_k^* = t_k$,

$$\begin{aligned} \frac{\|t_j + w\|}{\|w\|} &= \|t_j^* - (-w)^*\| \\ &\leq \|t_j^* - t_k^*\| + \|t_k^* - (-w)^*\| \\ &= \|t_j - t_k\| + \frac{\|t_k + w\|}{\|w\|}. \end{aligned}$$

Multiplying by $\|w\|$ gives $n_j - n_k \leq \|w\| \|t_j - t_k\|$, and exchanging j and k gives the absolute value. (This is Ptolemy's inequality for the four points t_j , t_k , $-w$, 0 .) The Cauchy-Schwarz route $2|\langle t_j - t_k, w \rangle| / (n_j + n_k) \leq 2\|w\| \|t_j - t_k\| / (n_j + n_k)$ proves the claim only when $n_j + n_k \geq 2$, which fails for translations such as $w = -\beta t_k$ with β near one; the inversion argument needs no such condition. The claim is sharp: for unit t_j, t_k with $\|t_j - t_k\| = 6/5$ and $w = -t_k + \frac{7}{20}(t_j + t_k) / \|t_j + t_k\|$ one has $n_k = 7/20$, $n_j = 5/4$, $\|w\| = 3/4$, and both sides equal $9/10$. Steps 4 and 5 together give the factor $\min\{1, \|w\|\}$.

Step 6 (consequences). With $\|w\| = \eta$ and $\|t_j - t_k\| \leq 2$ the chain gives $s_j - s_k \leq \eta \|t_j - t_k\| \leq 2\eta$. If the top-1 label of z changes from j to k , then $s_j \geq s_i$ and $\langle z, t'_k \rangle \geq \langle z, t'_i \rangle$ for all i , so the pair (j, k) satisfies the hypotheses and the original top-1 margin obeys $s_j - \max_{i \neq j} s_i \leq s_j - s_k \leq |n_j - n_k| \leq \max_{i, l} |n_i - n_l|$. Finally, if $\langle z, t'_j \rangle > 0$ and $s_j > s_k$, Step 1 forces $n_j - n_k > 0$. $\square$

Proof of Corollary 1. For the projected-gap translation, $p = (I - VV^\top)(\mu_I - \mu_T)$ with V an orthonormal basis of $\text{span}\{t_k - \mu_T\}$. Since $t_i - t_j = (t_i - \mu_T) - (t_j - \mu_T)$ lies in that span, $\langle p, t_i - t_j \rangle = 0$ for all i, j , and every multiple $w = cp$ inherits this. Now $\langle t_k, w \rangle = \langle t_k - t_1, w \rangle + \langle t_1, w \rangle = \langle t_1, w \rangle$ for every k , so $n_k^2 = 1 + 2\langle t_1, w \rangle + \|w\|^2$ is the same number n^2 for every class. By Proposition 1, $\langle z, t'_k \rangle = (s_k + \omega) / n$ is a strictly increasing affine function of s_k whose coefficients do not depend on k , so the set of maximizers, and in particular the hard label, is unchanged for every z . On a fixed set of draws the translated bank therefore produces identical hard labels, and the vote counts, the selected class, the one-sided confidence bound, the radius, and every accuracy computed from them coincide with those of the uncorrected bank. $\square$

Proof of Corollary 2. Let F be the set of draws whose argmax changes and, for $m \in F$, let j_m and $k_m \neq j_m$ be the original and translated winners. Then $s_{j_m} \geq s_{k_m}$ and $\langle z_m, t'_{j_m} \rangle \geq \langle z_m, t'_{k_m} \rangle$, so Proposition 2 gives $s_{j_m} - s_{k_m} \leq |n_{j_m} - n_{k_m}|$: every $m \in F$ is eligible and $|F| \leq E_F$. A draw that is eligible through the pair (j, k) has top-1 margin $s_j - \max_{i \neq j} s_i \leq s_j - s_k \leq |n_j - n_k| \leq \max_{i, l} |n_i - n_l|$, which gives the stated upper bound on E_F . Class i loses the votes of the draws in F with $j_m = i$ and gains those with $k_m = i$, so $v'_i \leq v_i + |\{m \in F : k_m = i\}| \leq v_i + E_F \leq k_0 + E_F$, and likewise $v'_i \geq v_i - E_F$. If $k_0 + E_F < k_{\min}$, then $\max_i v'_i < k_{\min}$. $\square$

Derivation of (6). With $t'_k = (t_k - \mu_T) / n_k$ and $z' = (z - \mu_I) / \|z - \mu_I\|$ the inner product equals $\langle z - \mu_I, t_k - \mu_T \rangle / (\|z - \mu_I\| n_k)$. The positive factor $1 / \|z - \mu_I\|$ is common to all k and does not affect the argmax, and expanding the numerator gives $s_k - \langle z, \mu_T \rangle - \langle \mu_I, t_k \rangle + \langle \mu_I, \mu_T \rangle$. The first two terms are exactly the text-only identity of Proposition 1 with $w = -\mu_T$, and the last term is common to all k ; the third term is a per-class bias that a common text translation cannot create, so the argument of Proposition 2 does not apply. An explicit instance in which the two-sided operator changes a label whose original margin exceeds every bound in (5) is verified in the released tests. $\square$

a) Per-class scales: The learned shared translation in its tangent form has $t'_k = (a_k t_k + v) / n_k$ with $a_k = 1 - \langle v, t_k \rangle > 0$ and $n_k = \|a_k t_k + v\|$. Repeating Step 1 with $a_k s_k$ in place of s_k gives, for a pair whose order changes, $a_j s_j - a_k s_k \leq (n_j - n_k) \langle z, t'_j \rangle \leq |n_j - n_k|$, and therefore $a_j (s_j - s_k) = a_j s_j - a_k s_k + (a_k - a_j) s_k \leq |n_j - n_k| + |a_j - a_k|$ because $|s_k| \leq 1$. This is the per-class-scale form of the registered fresh-noise prediction for that control.

APPENDIX C

ADDITIONAL TABLES AND PROTOCOL DETAILS

A. The complete audit family at the primary radius

Table IV lists every registered bank of the audit at $r = 0.25$: clean accuracy, both certified-accuracy metrics, the standard-metric contrast with its complete-family simultaneous band (Figure 3 shows both metrics), and eligibility under the saved

TABLE IV

COMPLETE 18-CANDIDATE FAMILY AT PIXEL-SPACE ℓ_2 RADIUS 0.25 (EQUAL-CELL MACRO OVER 12 CELLS, 3,300 POSITIONS; ACCURACIES IN PERCENT). STANDARD CA IS STANDARD COHEN CERTIFIED ACCURACY; ANCHORED CA ADDITIONALLY REQUIRES THE SELECTED SMOOTHED CLASS TO EQUAL THE RAW-CLEAN PREDICTION. THE STANDARD DELTA VERSUS NO CORRECTION CARRIES THE COMPLETE-FAMILY SIMULTANEOUS 95% BAND (COMMON HALF WIDTH 1.98 POINTS, 100,000 REPLICATES, 136 CONTRASTS). FLOAT GATE: HISTORICAL ELIGIBILITY FLAG AS SAVED; EXACT GATE: THE WRITTEN CLEAN-ACCURACY RULE RECOMPUTED WITH INTEGER COUNTS. † SAVED SELECTION; ‡ EXACT-COUNT SELECTION.

Candidate	Clean acc.	Std. CA	Anch. CA	Δ std. (points)	95% simultaneous band	Eligible (float rule)	Eligible (exact rule)
No correction	60.88	11.47	8.42	reference	–	yes	yes
Mean-centering (text) $c=0.25$	62.33	11.40	8.85	−0.07	[−2.05, +1.92]	no	no
Mean-centering (text) $c=0.5$	59.78	10.08	8.17	−1.38	[−3.37, +0.60]	no	no
Mean-centering (text) $c=1$	47.33	10.77	9.42	−0.70	[−2.68, +1.28]	no	no
GR-CLIP-style two-sided $c=1$	64.92	14.43	11.35	+2.97	[+0.98, +4.95]	no	no
Projected gap $c=0.25$	60.88	11.47	8.42	+0.00	[−1.98, +1.98]	yes	yes
Projected gap $c=0.5$	60.88	11.47	8.42	+0.00	[−1.98, +1.98]	yes	yes
Projected gap $c=1$	60.88	11.47	8.42	+0.00	[−1.98, +1.98]	yes	yes
Boundary-active step 0.0025†	61.08	11.30	8.43	−0.17	[−2.15, +1.82]	yes	yes
Boundary-active step 0.005	60.93	11.30	8.43	−0.17	[−2.15, +1.82]	no	yes
Boundary-active step 0.01	60.98	11.15	8.45	−0.32	[−2.30, +1.67]	no	yes
Boundary-active step 0.02	61.63	12.00	9.12	+0.53	[−1.45, +2.52]	no	yes
Boundary-active step 0.04‡	62.32	12.05	9.35	+0.58	[−1.40, +2.57]	no	yes
Noisy-margin step 0.0025	60.90	11.47	8.42	+0.00	[−1.98, +1.98]	no	yes
Noisy-margin step 0.005	60.90	11.47	8.42	+0.00	[−1.98, +1.98]	no	yes
Noisy-margin step 0.01	60.92	11.32	8.43	−0.15	[−2.13, +1.83]	no	yes
Noisy-margin step 0.02	60.37	11.32	8.43	−0.15	[−2.13, +1.83]	no	yes
Noisy-margin step 0.04	59.75	11.15	8.58	−0.32	[−2.30, +1.67]	no	no

floating-point and the exact-count readings of the clean gate (Section V-A).

B. Per-cell behavior of the two-sided operator

Table V lists clean accuracy and both certified-accuracy metrics at $r = 0.25$ for the uncorrected model and for the GR-CLIP-style two-sided centering in every cell. The clean-gate rejection is driven by a single EuroSAT cell; the standard-metric gain is +0.60 points on the CIFAR-100 cells and +5.33 points on the EuroSAT cells, which is why the item-pooled gain (+1.03 points) is smaller than the equal-cell gain (+2.97 points).

C. Registered extension design, regime and contrasts

Tables VI, VII and VIII give the extension’s registered design, the uncorrected model in every cell, and every registered contrast with its band (Section VII).

D. Fresh-noise diagnostic: flip budget and taxonomy

Table IX reports, per cell and bank, the registered flip-budget partition (items already at the confirmation threshold, items for which the threshold is unattainable on the recorded draws, and undetermined items) and the per-draw taxonomy of label changes against ground truth (useful wrong-to-correct, harmful correct-to-wrong, wrong-to-wrong, unchanged), with 0 recorded exceptions to the decision-change inequality (prediction P1). The taxonomy is descriptive; it identifies no cause. The diagnostic’s other registered predictions are as follows: P2 (the projected-gap banks) recorded an orthogonality residual of at most 7.65×10^{-8} and a normalizer mismatch of at most 2.20×10^{-7} , and all 35 of their float-level flips satisfied the inequality (0 exceptions); P4a (Gaussian calibration of the noisy margin) returned *pass*, with a mean absolute calibration error of at most 0.0028; P5 (a class’s noisy vote share rises with its alignment to the mean noisy feature) returned *pass*, with

Spearman correlations from +0.62 to +0.97 across the four cells; P6 (descriptive margin anatomy) gives the items already at threshold in Table IX, with the per-draw margin quantiles in the released outputs; and P8 (the extension’s controls were frozen and hash-bound before the fresh noise) is procedural and has no numerical outcome.

E. Exact-count clean-gate rule

The registered rule declares a candidate eligible when its macro clean-accuracy drop over the 12 cells is at most one point and no cell drops by more than two points, with drops measured against the same-cell uncorrected model. For a cell with n items and correct counts c_{ref} and c_{cand} , the drop in points is $100(c_{\text{ref}} - c_{\text{cand}})/n$, an exact rational number. The implementation evaluated $100(a_{\text{ref}} - a_{\text{cand}}) \leq 2$ with floating-point accuracies, which represents a one-image drop in a 50-image cell as 2.0000000000000018. The released adjudication script recomputes the rule with exact rational arithmetic, reports both readings, and leaves every saved file unchanged.

F. Compute and environment

The immutable audit ran on one consumer GPU (an RTX 4090) under a pinned runtime recorded in the artifact (Python 3.12, PyTorch 2.11 with CUDA 13, OpenCLIP 3.3.0) and took 14.4 GPU-hours for its 243,302,400 confirmation votes. Encoders ran in float32 with reduced-precision matrix modes disabled. The positive-control pilot trained its adapters and certified its items on a rented GPU; its confirmation stage used the same budget per item. The complete-family analysis (100,000 replicates), the reconstruction scripts and every table in this paper run on a CPU from the released statistics and recorded outputs without model or dataset downloads. No experiment was retuned or repeated in response to an inspected outcome: the same-item re-certification of Section VIII was registered as such before its sampling, and one partially

TABLE V

PER-CELL CLEAN ACCURACY, STANDARD CERTIFIED ACCURACY (STD CA) AND ANCHORED CERTIFIED ACCURACY (ANCH. CA) AT RADIUS 0.25 FOR NO CORRECTION AND THE GR-CLIP-STYLE TWO-SIDED CANDIDATE (COEFFICIENT 1). DROP IS THE CLEAN-ACCURACY DECREASE IN POINTS COMPUTED FROM INTEGER COUNTS (NEGATIVE VALUES ARE IMPROVEMENTS); THE WORST CELL IS ViT-L/14, EUROSAT, FOLD 1 AT 4.00 POINTS (2 IMAGES) AND 9 OF 12 CELLS IMPROVE. COLUMN MEANS EQUAL THE EQUAL-CELL MACROS (11.47, 14.43, 8.42, 11.35).

Model	Dataset	Fold	No corr. clean	GR-CLIP clean	Drop (pts)	No corr. Std CA	GR-CLIP Std CA	No corr. Anch. CA	GR-CLIP Anch. CA
ViT-B/32	CIFAR-100	0	64.60	65.40	−0.80	7.40	7.60	6.40	6.80
ViT-B/32	CIFAR-100	1	59.40	63.60	−4.20	7.00	7.60	6.60	7.60
ViT-B/32	CIFAR-100	2	63.40	63.20	+0.20	9.20	8.80	7.40	7.80
ViT-B/32	EuroSAT	0	50.00	62.00	−12.00	4.00	12.00	2.00	8.00
ViT-B/32	EuroSAT	1	42.00	48.00	−6.00	6.00	14.00	4.00	8.00
ViT-B/32	EuroSAT	2	46.00	58.00	−12.00	2.00	12.00	2.00	6.00
ViT-L/14	CIFAR-100	0	73.00	79.60	−6.60	24.60	27.40	21.60	26.00
ViT-L/14	CIFAR-100	1	74.80	80.20	−5.40	21.20	22.20	19.00	20.80
ViT-L/14	CIFAR-100	2	75.40	79.00	−3.60	26.20	25.60	24.00	25.20
ViT-L/14	EuroSAT	0	58.00	60.00	−2.00	12.00	14.00	4.00	8.00
ViT-L/14	EuroSAT	1	66.00	62.00	+4.00	8.00	10.00	2.00	2.00
ViT-L/14	EuroSAT	2	58.00	58.00	+0.00	10.00	12.00	2.00	10.00

TABLE VI

REGISTERED EXTENSION: CELLS, ITEMS PER CELL, NOISE SCALES, PROTOTYPE BANKS (FROZEN GRID PLUS LEARNED CONTROLS), CONTROLS, PRIMARY ESTIMAND, AND REGISTRATION STATUS WITH THE FIRST TWELVE HEX CHARACTERS OF EACH REGISTRATION SHA-256.

Study	Cells	Items per cell	σ	Banks	Controls	Primary estimand	Status
Extension A	14 (12 primary)	500 (CIFAR-100, CIFAR-10); (EuroSAT)	0.12, 0.25, 150 0.5	22 (18 + 4)	4	paired macro standard certified accuracy at $r = \text{sigma}$ versus no_correction over the balanced 12 extension cells (2 models x 3 datasets x sigma in {0.12,0.5}), equal weight per cell	preregistered extension (9802bf3e5f97)
Extension B	18	500 (CIFAR-100, CIFAR-10); (EuroSAT)	0.25, 0.5 150	22 (18 + 4)	4	paired macro standard certified accuracy at $r = \text{sigma}$ versus no_correction over all 18 backbone-extension cells (3 models x 3 datasets x sigma in {0.25,0.5}), equal weight per cell	preregistered extension (3fcdade664a)
Fresh-noise diagnostic	4	1,000 (CIFAR-100); 300 (EuroSAT)	0.25	22 (18 + 4)	4	8 conformance predictions (P1, P2, P4a, P5, P6, P7, P8, P9); no certificate is issued	preregistered diagnostic (96fd42f268d9)

completed follow-up job was discarded after a hardware fault and recomputed from its first item (Appendix E); the extension of Section VII was registered before any of its conditions were sampled, and its plan, from measured throughput, was 23 GPU-hours on the audit's GPU. The follow-up's sampling ran for 15.0 hours on seven RTX 5090 GPUs of one rented host, after an eighth developed a hardware fault, and for 15.3 hours on one RTX 5090 of a second host (Appendix E); its registered analysis runs on a CPU.

TABLE VII

NO CORRECTION IN EVERY REGISTERED CELL OF EXTENSIONS A AND B: ITEMS, CLEAN ACCURACY, SMOOTHED ACCURACY, STANDARD CERTIFIED ACCURACY AND CERTIFIED-WRONG RATE AT $r = \sigma$, ABSTENTION, AND THE TOP-CLASS SHARE OF THE SMOOTHED PREDICTIONS. † MARKS THE TWO CIFAR-10 $\sigma = 0.25$ BRIDGE CELLS, WHICH ARE REPORTED BUT EXCLUDED FROM THE PRIMARY MACRO. VALUES ARE PRIMARY-VERIFIED (APPENDIX E).

Model	Dataset	σ	n	Clean	Smoothed	Std CA ($r = \sigma$)	Cert.-wrong ($r = \sigma$)	Abstain	Top-class share
<i>Extension A</i>									
ViT-B/32	CIFAR-10	0.12	500	90.00	76.60	58.40	6.00	8.80	15.20
ViT-B/32	CIFAR-10	0.25 [†]	500	90.00	45.20	24.20	7.80	20.20	30.00
ViT-B/32	CIFAR-10	0.5	500	90.00	23.00	7.20	27.80	21.40	69.80
ViT-B/32	CIFAR-100	0.12	500	61.00	40.00	26.00	12.80	25.60	7.80
ViT-B/32	CIFAR-100	0.5	500	61.00	6.80	3.00	31.40	25.20	56.40
ViT-B/32	EuroSAT	0.12	150	47.33	9.33	5.33	52.67	6.00	50.00
ViT-B/32	EuroSAT	0.5	150	47.33	9.33	8.67	69.33	0.67	97.33
ViT-L/14	CIFAR-10	0.12	500	96.20	90.00	81.80	2.40	4.40	11.60
ViT-L/14	CIFAR-10	0.25 [†]	500	96.20	71.60	51.00	4.80	12.40	16.20
ViT-L/14	CIFAR-10	0.5	500	96.20	41.20	16.40	9.20	23.80	26.40
ViT-L/14	CIFAR-100	0.12	500	72.40	63.00	48.20	8.00	16.20	5.00
ViT-L/14	CIFAR-100	0.5	500	72.40	14.60	6.80	12.00	36.20	10.80
ViT-L/14	EuroSAT	0.12	150	66.00	26.67	16.00	42.00	6.67	59.33
ViT-L/14	EuroSAT	0.5	150	66.00	10.00	9.33	89.33	0.00	100.00
<i>Extension B</i>									
RN50	CIFAR-10	0.25	500	73.60	17.60	8.20	31.80	13.40	53.20
RN50	CIFAR-10	0.5	500	73.60	12.20	9.60	55.80	5.80	50.60
RN50	CIFAR-100	0.25	500	38.80	3.80	1.00	41.80	21.20	61.80
RN50	CIFAR-100	0.5	500	38.80	1.00	1.00	65.00	4.20	61.80
RN50	EuroSAT	0.25	150	37.33	10.00	9.33	47.33	7.33	65.33
RN50	EuroSAT	0.5	150	37.33	16.00	10.67	42.00	2.67	50.00
ViT-B/16	CIFAR-10	0.25	500	91.20	47.20	24.00	9.20	23.60	24.40
ViT-B/16	CIFAR-10	0.5	500	91.20	26.20	6.20	8.40	35.00	35.00
ViT-B/16	CIFAR-100	0.25	500	64.60	18.60	9.20	38.80	17.20	54.40
ViT-B/16	CIFAR-100	0.5	500	64.60	5.40	2.20	35.20	17.20	64.40
ViT-B/16	EuroSAT	0.25	150	56.67	12.67	4.67	45.33	15.33	80.67
ViT-B/16	EuroSAT	0.5	150	56.67	14.67	5.33	22.00	29.33	43.33
ViT-B/32 (LAION-2B)	CIFAR-10	0.25	500	93.60	41.60	22.80	14.00	17.20	29.80
ViT-B/32 (LAION-2B)	CIFAR-10	0.5	500	93.60	15.40	7.00	26.20	15.80	70.80
ViT-B/32 (LAION-2B)	CIFAR-100	0.25	500	75.40	13.20	6.80	37.20	15.40	57.00
ViT-B/32 (LAION-2B)	CIFAR-100	0.5	500	75.40	5.00	1.80	50.00	13.60	78.60
ViT-B/32 (LAION-2B)	EuroSAT	0.25	150	50.00	12.67	5.33	43.33	12.00	70.67
ViT-B/32 (LAION-2B)	EuroSAT	0.5	150	50.00	10.00	8.00	29.33	9.33	90.00

TABLE VIII

REGISTERED CONTRASTS VERSUS NO CORRECTION IN THE EXTENSION: EQUAL-CELL MACRO CHANGE OF STANDARD CERTIFIED ACCURACY AT $r = \sigma$ OVER THE PRIMARY CELLS (POINTS), THE REGISTERED SHARED-ITEM CLASS-STRATIFIED SIMULTANEOUS 95% BAND, THE EQUAL-CELL CLEAN-ACCURACY CHANGE (POINTS), WHETHER THE BAND EXCLUDES ZERO, AND WHETHER A GRID BANK'S CHANGE LIES WITHIN THE REGISTERED ± 0.5 -POINT REPORTING REGION (PREDICTION X1; A REPORTING REGION, NOT AN EQUIVALENCE MARGIN). SEVENTEEN FROZEN-GRID BANKS PRECEDE THE FOUR CONTROLS; THE CONTROLS ARE SUPERVISED WITH 20 LABELS PER CLASS FROM THE DISJOINT CONTROL-TRAINING SPLIT. COMMON HALF WIDTHS: EXTENSION A: 1.84 POINTS; EXTENSION B: 1.56 POINTS. VALUES ARE PRIMARY-VERIFIED (APPENDIX E).

Candidate	Δ std. CA (points)	95% simultaneous band	Δ clean (points)	Band excludes 0	Within ± 0.5
<i>Extension A</i>					
Mean-centering (text) $c=0.25$	-0.14	[-1.98, +1.70]	+1.67	no	yes
Mean-centering (text) $c=0.5$	-0.51	[-2.35, +1.33]	-1.47	no	no
Mean-centering (text) $c=1$	-2.56	[-4.40, -0.72]	-12.31	yes	no
Two-sided centering $c=1$	+0.32	[-1.52, +2.16]	+4.66	no	yes
Projected gap $c=0.25$	+0.00	[-1.84, +1.84]	+0.00	no	yes
Projected gap $c=0.5$	+0.00	[-1.84, +1.84]	+0.00	no	yes
Projected gap $c=1$	+0.00	[-1.84, +1.84]	+0.00	no	yes
Boundary-active step 0.0025	+0.06	[-1.78, +1.89]	-0.03	no	yes
Boundary-active step 0.005	+0.07	[-1.77, +1.91]	+0.00	no	yes
Boundary-active step 0.01	+0.07	[-1.77, +1.91]	+0.08	no	yes
Boundary-active step 0.02	+0.12	[-1.72, +1.96]	-0.26	no	yes
Boundary-active step 0.04	+0.17	[-1.67, +2.01]	-0.62	no	yes
Noisy-margin step 0.0025	-0.03	[-1.87, +1.81]	-0.24	no	yes
Noisy-margin step 0.005	+0.04	[-1.80, +1.88]	-0.27	no	yes
Noisy-margin step 0.01	+0.11	[-1.73, +1.95]	-0.54	no	yes
Noisy-margin step 0.02	+0.11	[-1.73, +1.94]	-0.83	no	yes
Noisy-margin step 0.04	+0.32	[-1.52, +2.16]	-1.69	no	yes
Learned shared translation (tangent)	-0.02	[-1.86, +1.82]	+1.29	no	-
Learned shared translation (pure)	-3.26	[-5.10, -1.42]	-11.46	yes	-
Noisy class mean (supervised)	+9.12	[+7.28, +10.96]	-39.44	yes	-
Rank-8 tangent adapter (supervised)	+15.09	[+13.26, +16.93]	+10.22	yes	-
<i>Extension B</i>					
Mean-centering (text) $c=0.25$	+0.26	[-1.31, +1.82]	-0.15	no	yes
Mean-centering (text) $c=0.5$	+0.75	[-0.81, +2.31]	-2.02	no	no
Mean-centering (text) $c=1$	+0.61	[-0.96, +2.17]	-7.67	no	no
Two-sided centering $c=1$	+0.42	[-1.14, +1.98]	+3.76	no	yes
Projected gap $c=0.25$	+0.00	[-1.56, +1.56]	+0.00	no	yes
Projected gap $c=0.5$	+0.00	[-1.56, +1.56]	+0.00	no	yes
Projected gap $c=1$	+0.00	[-1.56, +1.56]	+0.00	no	yes
Boundary-active step 0.0025	-0.02	[-1.59, +1.54]	+0.07	no	yes
Boundary-active step 0.005	-0.02	[-1.58, +1.54]	+0.09	no	yes
Boundary-active step 0.01	-0.01	[-1.58, +1.55]	+0.13	no	yes
Boundary-active step 0.02	-0.15	[-1.71, +1.41]	+0.13	no	yes
Boundary-active step 0.04	-0.13	[-1.69, +1.43]	-0.15	no	yes
Noisy-margin step 0.0025	+0.04	[-1.53, +1.60]	+0.01	no	yes
Noisy-margin step 0.005	+0.03	[-1.54, +1.59]	-0.01	no	yes
Noisy-margin step 0.01	-0.01	[-1.57, +1.56]	+0.04	no	yes
Noisy-margin step 0.02	+0.05	[-1.51, +1.61]	-0.63	no	yes
Noisy-margin step 0.04	+0.01	[-1.55, +1.57]	-1.25	no	yes
Learned shared translation (tangent)	+0.96	[-0.60, +2.52]	-1.85	no	-
Learned shared translation (pure)	-0.08	[-1.64, +1.48]	-9.79	no	-
Noisy class mean (supervised)	+7.34	[+5.78, +8.90]	-42.29	yes	-
Rank-8 tangent adapter (supervised)	+13.67	[+12.11, +15.24]	+12.88	yes	-

TABLE IX

FRESH-NOISE DIAGNOSTIC PER REGISTERED CELL FOR FOUR BANKS ($k_{\min} = 456$ OF 512 DRAWS). P9: DRAW-LEVEL WINNER MOVEMENTS AGAINST GROUND TRUTH (USEFUL = WRONG-TO-CORRECT, HARMFUL = CORRECT-TO-WRONG). P7: ITEMS WHOSE IDENTITY-BANK TOP COUNT ALREADY REACHES $k_{\min}$, ITEMS FOR WHICH THE THRESHOLD IS UNATTAINABLE ON THESE DRAWS UNDER THE CANDIDATE'S FLIP BUDGET, AND UNDETERMINED ITEMS; N/A WHERE THE CONTAINMENT OPERATOR CLASS DOES NOT APPLY. NO CERTIFICATE IS ISSUED.

Bank	P9 useful	P9 harmful	P9 wrong-to-wrong	P9 unchanged	P7 at $k_{\min}$	P7 unattainable	P7 undetermined
<i>ViT-B/32, CIFAR-100</i> (1,000 items, 512 draws, 311 items already at $k_{\min}$)							
Mean-centering (text) $c=1$	16,834	45,497	266,434	183,235	311	0	689
Two-sided centering $c=1$	16,712	19,282	128,398	347,608	n/a	n/a	n/a
Boundary-active step 0.04	3,057	2,253	24,736	481,954	311	101	588
Rank-8 tangent adapter (supervised)	73,010	25,318	296,526	117,146	n/a	n/a	n/a
<i>ViT-B/32, EuroSAT</i> (300 items, 512 draws, 72 items already at $k_{\min}$)							
Mean-centering (text) $c=1$	15,115	17,666	117,493	3,326	72	0	228
Two-sided centering $c=1$	11,464	8,032	55,734	78,370	n/a	n/a	n/a
Boundary-active step 0.04	1,595	824	7,010	144,171	72	0	228
Rank-8 tangent adapter (supervised)	80,601	8,452	45,227	19,320	n/a	n/a	n/a
<i>ViT-L/14, CIFAR-100</i> (1,000 items, 512 draws, 338 items already at $k_{\min}$)							
Mean-centering (text) $c=1$	18,190	63,401	165,415	264,994	338	0	662
Two-sided centering $c=1$	20,363	30,122	127,393	334,122	n/a	n/a	n/a
Boundary-active step 0.04	2,438	2,214	15,056	492,292	338	259	403
Rank-8 tangent adapter (supervised)	63,781	27,094	158,706	262,419	n/a	n/a	n/a
<i>ViT-L/14, EuroSAT</i> (300 items, 512 draws, 226 items already at $k_{\min}$)							
Mean-centering (text) $c=1$	12,383	14,882	88,616	37,719	226	0	74
Two-sided centering $c=1$	15,054	6,582	32,895	99,069	n/a	n/a	n/a
Boundary-active step 0.04	1,017	1,882	5,117	145,584	226	0	74
Rank-8 tangent adapter (supervised)	82,955	4,849	48,981	16,815	n/a	n/a	n/a

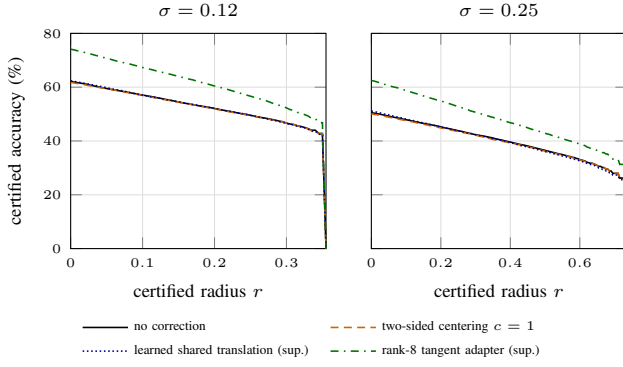


Fig. 5. Descriptive, not registered: Follow-up B standard certified accuracy against the certified radius, pooled over all cells at each noise level, for the uncorrected model, two-sided centering and two supervised controls. The curves use the registered certification function on the custody-verified vote counts, and at every registered radius they equal the registered analysis output. The four banks were fixed before any outcome existed.

APPENDIX D

FOLLOW-UP: DESIGN, CONTROLS, REGIME AND RADII

Table X summarizes the three registrations, Table XI gives both estimands and the adjusted classes of the pooled primary family, Table XII gives the secondary dataset-specific families, Table XIII gives the supervised controls and the text/image decomposition at every coefficient, and Table XIV gives the step response of the proposal objectives. The observed paired discordance stands next to every interval of Tables XI and XII (for Table XIII see Appendix E); -0.00 marks a negative value that rounds to zero.

a) Supervised controls (Q5): In the pooled Follow-up B families the supervised controls change standard certified accuracy at $r = \sigma$ by $+9.90$ points at $\sigma = 0.12$ and $+8.97$ at $\sigma = 0.25$ for the rank-8 tangent adapter (Holm-adjusted direction positive and positive), -0.06 and -0.03 for the learned shared translation (unresolved, unresolved), $+9.35$ and $+8.03$ for the few-shot prompt bank (positive, positive), and $+6.08$ and $+5.68$ for the noisy class-mean bank (positive, positive). On the audit’s items (Follow-up A) they change it by $+4.61$ (rank-8 adapter, positive), $+0.00$ (learned shared translation, unresolved), $+6.88$ (few-shot prompt bank, positive) and $+1.06$ points (noisy class means, unresolved). Over the Follow-up B cells their macro clean accuracy changes by $+12.26$, $+0.61$, $+11.75$ and -20.74 points in the same order, and the registered clean rule flags the noisy class-mean bank in 13 of the 18 cells and the learned shared translation in 3; as for the positive control (Section VI), the certified gains of the controls that change prototypes class by class are not isolated from ordinary changes in clean accuracy. The few-shot prompt bank fits four shared context vectors through the frozen text encoder; it is inspired by PromptSmooth [14] and is not a reproduction of it.

b) Regime of the uncorrected model (Q7): Across the Follow-up B cells the uncorrected model’s smoothed accuracy ranges from 10.40% to 99.37% at $\sigma = 0.12$ and from 11.10%

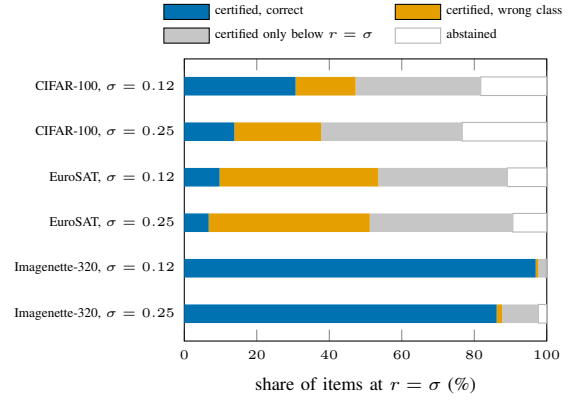


Fig. 6. How the uncorrected model’s smoothed decisions split at $r = \sigma$ on the fresh Follow-up B items, per dataset and noise level, pooled over backbones (from the registered cells table, Q7): certified and correct, certified for a wrong class, certified only at a smaller radius, or abstained.

to 96.70% at $\sigma = 0.25$, and its standard certified accuracy at $r = \sigma$ from 5.50% to 98.77% and from 5.20% to 92.80%. Figure 6 shows how its decisions split at $r = \sigma$, including certificates for a wrong class. The per-cell values, with the top-class share and the abstention rate, are in the registered analysis output.

c) Orientation against published radii (Q8): On Imagenette at $\sigma = 0.25$ the uncorrected model’s standard certified accuracy is 84.30%, 92.80% and 81.43% at $r = 0.25$ and 72.37%, 85.03% and 69.40% at $r = 0.5$ (OpenAI ViT-B/32, OpenAI ViT-L/14 and OpenCLIP ViT-B/32 trained on LAION-2B). Nirala et al. report certified accuracy curves for zero-shot CLIP on ImageNet, including ViT-B/32 backbones from OpenAI and OpenCLIP [3]. They certify after CLIP’s channel standardization, whose standard deviations lie between 0.26 and 0.28, so the same pixel perturbation has an ℓ_2 norm 3.6 to 3.8 times larger in their coordinates than in ours (Section II); their class set and sampling budget also differ. The registration makes no comparative claim, and neither do we.

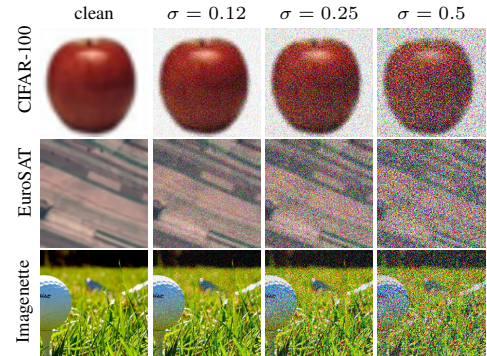


Fig. 7. One development image per dataset in the 224×224 model-input coordinates of Section II (before channel standardization), clean and with one fixed Gaussian draw at each σ . The display clips to $[0, 1]$; the classifier sees unclipped samples. A fixed rule (the first development item of a fixed class), not the outcome, chose the images.

TABLE X

REGISTERED FOLLOW-UP (REGISTERED BEFORE SAMPLING; REGISTRATION DIGESTS D1FBE0A89484, 9A1A5C6E1A46 AND 2359B499C96C): CELLS, BACKBONES, DATASETS WITH ITEMS PER CELL, NOISE LEVELS, BANKS, SELECTION AND CONFIRMATION DRAWS, THE PER-EXAMPLE ERROR LEVEL α OF THE ONE-SIDED CLOPPER–PEARSON BOUND (CONFIDENCE $1 - \alpha$), REGISTERED QUESTIONS, AND REGISTERED POWER (MINIMUM DETECTABLE EFFECT IN POINTS AT PAIRED DISCORDANCE 0.10). INFERENCE: CLASS-STRATIFIED SHARED-ITEM BOOTSTRAP WITH 100,000 REPLICATES, HOLM WITHIN EACH FAMILY, SMALLEST EFFECT OF INTEREST 2 POINTS.

Study	Cells	Backbones	Datasets (items per cell)	σ	Banks	Draws n_0 / n	α	Questions	Power
Follow-up A	12	ViT-B/32, ViT-L/14	CIFAR-100 validation folds (500); EuroSAT validation folds (50)	0.25	36	128 / 4,096	0.001	Q1–Q5	MDE 2.7
Follow-up B	18	ViT-B/32, ViT-L/14, ViT-B/32 (LAION-2B)	CIFAR-100 test (3,000); EuroSAT sealed holdout (1,000); Imagenette-320 validation (3,000)	0.12, 0.25	36	128 / 4,096	0.001	Q2–Q8	pooled MDE 1.3 (power 0.997 at 2); EuroSAT MDE 3.47
Follow-up C	6	ViT-B/32, ViT-L/14	CIFAR-100 test (100); EuroSAT sealed holdout (100); Imagenette-320 validation (100)	0.25	3	128 / 100,000 (prefix 4,096)	0.001	Q9	descriptive

TABLE XI

FOLLOW-UP B (FRESH ITEMS): CHANGE IN STANDARD CERTIFIED ACCURACY AT $r = \sigma$ VERSUS NO CORRECTION, IN POINTS, POOLED OVER DATASETS AND BACKBONES, WITH UNADJUSTED 95% BOOTSTRAP INTERVALS, ITEM-POOLED (THE REGISTERED PRIMARY ESTIMAND) AND EQUAL-CELL (THE AUDIT’S ESTIMAND, REPORTED ALONGSIDE). FOR THE FIVE CONTRASTS OF THE REGISTERED PRIMARY FAMILY, THE HOLM-ADJUSTED DIRECTION (POS., NEG., UNRES. = UNRESOLVED) AND MAGNITUDE CLASS AGAINST THE 2-POINT SMALLEST EFFECT OF INTEREST FOLLOW (GAIN, LOSS, EQUIV. = PRACTICALLY EQUIVALENT, INCONCL. = INCONCLUSIVE); AN UNADJUSTED INTERVAL CAN EXCLUDE ZERO WHILE THE ADJUSTED DIRECTION IS UNRESOLVED. d : THE OBSERVED PAIRED DISCORDANCE (%), THE SHARE OF POSITIONS WHOSE CERTIFIED-CORRECT OUTCOME DIFFERS FROM NO CORRECTION’S), REPORTED NEXT TO EVERY INTERVAL AS REGISTERED. THE LAST THREE ROWS ARE REFERENCE CONTRASTS.

Bank	Item-pooled [95% CI]	Equal-cell [95% CI]	d (%)	Holm dir.	Class
$\sigma = 0.12$					
Two-sided centering $c = 1$	−0.18 [−0.50, +0.13]	−0.24 [−0.64, +0.16]	6.5	unres.	equiv.
its text half	−0.68 [−0.93, −0.42]	−0.55 [−0.82, −0.28]	3.6	neg.	equiv.
its image half	+0.30 [−0.00, +0.61]	+0.27 [−0.13, +0.67]	6.4	unres.	equiv.
Boundary-active proposal, step 0.16	+0.08 [−0.14, +0.29]	+0.34 [+0.05, +0.64]	2.6	unres.	equiv.
Noisy-margin proposal, step 0.16	+0.50 [+0.31, +0.68]	+1.21 [+0.88, +1.55]	2.0	pos.	equiv.
Projected gap $c = 1$	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	0.0	–	–
Rank-8 adapter (supervised)	+9.90 [+9.36, +10.43]	+18.33 [+17.45, +19.21]	16.1	–	–
Learned shared translation (supervised)	−0.06 [−0.32, +0.20]	+0.65 [+0.31, +1.00]	4.1	–	–
$\sigma = 0.25$					
Two-sided centering $c = 1$	−0.35 [−0.65, −0.05]	+0.14 [−0.21, +0.48]	5.9	unres.	equiv.
its text half	−0.93 [−1.14, −0.72]	−0.85 [−1.07, −0.63]	2.5	neg.	equiv.
its image half	+0.21 [−0.09, +0.52]	+0.77 [+0.41, +1.12]	6.3	unres.	equiv.
Boundary-active proposal, step 0.16	−1.12 [−1.37, −0.88]	−0.99 [−1.31, −0.68]	3.6	neg.	equiv.
Noisy-margin proposal, step 0.16	−0.10 [−0.27, +0.08]	−0.16 [−0.46, +0.14]	1.9	unres.	equiv.
Projected gap $c = 1$	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	0.0	–	–
Rank-8 adapter (supervised)	+8.97 [+8.40, +9.54]	+14.83 [+13.93, +15.72]	15.5	–	–
Learned shared translation (supervised)	−0.03 [−0.29, +0.22]	+0.54 [+0.16, +0.91]	4.2	–	–

d) Inference procedure: The primary estimand is the item-pooled paired difference in standard certified accuracy at $r = \sigma$, bank minus no correction, in points. Uncertainty comes from a class-stratified, shared-item paired bootstrap with 100,000 replicates and a registered seed. The resampling unit is the unique evaluation image: in each replicate its multiplicity is drawn once within its (dataset, class) stratum and shared by every backbone and noise level in which the image appears and by every bank and radius, so the dependence between backbones evaluated on the same image is kept. Each replicate Δ_b adds to the point estimate the resampled deviation rescaled by $\sqrt{m/(m-1)}$ for a stratum of m images (Rao–Wu). With $\Pr^*$ the share of replicates and the margin $\delta = 2$ points, the direction test uses $p = 2 \min\{\Pr^*(\Delta_b \leq 0), \Pr^*(\Delta_b \geq 0)\}$; the magnitude tests use $p = \Pr^*(\Delta_b \leq \delta)$ for a gain of at least δ , $p = \Pr^*(\Delta_b \geq -\delta)$ for a loss of at least δ , and the two one-sided tests $p = \max\{\Pr^*(\Delta_b \leq -\delta), \Pr^*(\Delta_b \geq \delta)\}$ for the

class practically equivalent; a contrast is a gain, else a loss, else practically equivalent, else inconclusive. Holm’s procedure acts within each family (the pooled family at each σ , which decides Q6, and each dataset at each σ), separately for each of the four tests; it adjusts decisions only, so the reported 95% percentile intervals are unadjusted. The four supervised controls form their own family against no correction, and a Romano–Wolf step-down max- t procedure covers every non-identity contrast as a secondary family.

TABLE XII

SECONDARY, DATASET-SPECIFIC FOLLOW-UP B FAMILIES (REGISTERED, NOT ADJUDICATING): CHANGE IN STANDARD CERTIFIED ACCURACY AT $r = \sigma$ VERSUS NO CORRECTION FOR THE FIVE PRIMARY CONTRASTS, POOLED OVER THE BACKBONES OF EACH DATASET, WITH THE UNADJUSTED 95% INTERVAL, THE HOLM-ADJUSTED MAGNITUDE CLASS WITHIN THE FAMILY, AND THE OBSERVED PAIRED DISCORDANCE d (%), THE SHARE OF POSITIONS WHOSE CERTIFIED-CORRECT OUTCOME DIFFERS FROM NO CORRECTION'S), REPORTED NEXT TO EVERY INTERVAL AS REGISTERED. HEADERS GIVE EACH FAMILY'S REGISTERED POWER LABEL. ADJUDICATION USES THE POOLED FAMILIES (FIGURE 4).

Bank	σ	CIFAR-100 test (powered)	EuroSAT sealed holdout (underpowered)	Imagenette-320 validation (powered)
Two-sided centering $c = 1$	0.12	+0.29 [−0.31,+0.89] equiv. $d=9.3$	−0.43 [−1.44,+0.57] equiv. $d=12.5$	−0.57 [−0.86,−0.29] equiv. $d=1.8$
	0.25	+0.23 [−0.21,+0.67] equiv. $d=4.8$	+1.87 [+1.06,+2.67] inconcl. $d=10.8$	−1.68 [−2.15,−1.21] inconcl. $d=5.3$
its text half	0.12	−1.23 [−1.75,−0.71] equiv. $d=6.4$	−0.10 [−0.67,+0.47] equiv. $d=3.1$	−0.31 [−0.52,−0.11] equiv. $d=0.9$
	0.25	−0.47 [−0.82,−0.12] equiv. $d=3.0$	−0.57 [−1.04,−0.10] equiv. $d=1.6$	−1.51 [−1.83,−1.20] equiv. $d=2.3$
its image half	0.12	+1.29 [+0.71,+1.87] equiv. $d=8.7$	+0.17 [−0.84,+1.17] equiv. $d=13.2$	−0.63 [−0.91,−0.36] equiv. $d=1.8$
	0.25	+0.53 [+0.08,+0.99] equiv. $d=5.0$	+2.70 [+1.86,+3.54] inconcl. $d=11.8$	−0.93 [−1.42,−0.44] equiv. $d=5.9$
Boundary-active proposal, step 0.16	0.12	+0.27 [−0.08,+0.62] equiv. $d=3.0$	+1.27 [+0.50,+2.04] inconcl. $d=5.4$	−0.51 [−0.78,−0.26] equiv. $d=1.3$
	0.25	+0.22 [−0.05,+0.49] equiv. $d=1.7$	−0.53 [−1.30,+0.24] equiv. $d=7.3$	−2.67 [−3.11,−2.22] loss $d=4.3$
Noisy-margin proposal, step 0.16	0.12	−0.11 [−0.37,+0.15] equiv. $d=1.6$	+3.73 [+2.76,+4.70] gain $d=8.3$	+0.02 [−0.09,+0.13] equiv. $d=0.3$
	0.25	+0.13 [−0.07,+0.34] equiv. $d=1.0$	−0.37 [−1.24,+0.50] equiv. $d=7.6$	−0.23 [−0.43,−0.04] equiv. $d=0.9$

TABLE XIII

REGISTERED FOLLOW-UP OUTPUTS SUMMARIZED IN THE TEXT, ITEM-POOLED AT $r = \sigma$ (POINTS). TOP (Q5): EACH SUPERVISED CONTROL AGAINST NO CORRECTION, WITH ITS HOLM-ADJUSTED DIRECTION, AND AGAINST TWO-SIDED CENTERING $c = 1$ WITH THE UNADJUSTED 95% INTERVAL. BOTTOM (Q2): TWO-SIDED CENTERING, ITS TEXT HALF AND ITS IMAGE HALF AT EVERY REGISTERED COEFFICIENT c , AND THEIR INTERACTION (TWO-SIDED MINUS BOTH HALVES) WITH ITS UNADJUSTED 95% INTERVAL.

Study	Control	vs. no correction	Holm dir.	vs. two-sided [95% CI]
Follow-up A, $\sigma = 0.25$	Rank-8 tangent adapter	+4.61	pos.	+3.64 [+2.32, +4.95]
	Learned shared translation	+0.00	unres.	−0.97 [−1.66, −0.28]
	Few-shot prompt bank	+6.88	pos.	+5.91 [+4.65, +7.20]
	Noisy class-mean bank	+1.06	unres.	+0.09 [−1.38, +1.57]
Follow-up B, $\sigma = 0.12$	Rank-8 tangent adapter	+9.90	pos.	+10.08 [+9.51, +10.64]
	Learned shared translation	−0.06	unres.	+0.12 [−0.19, +0.44]
	Few-shot prompt bank	+9.35	pos.	+9.53 [+9.02, +10.05]
	Noisy class-mean bank	+6.08	pos.	+6.26 [+5.64, +6.88]
Follow-up B, $\sigma = 0.25$	Rank-8 tangent adapter	+8.97	pos.	+9.32 [+8.73, +9.91]
	Learned shared translation	−0.03	unres.	+0.32 [+0.01, +0.63]
	Few-shot prompt bank	+8.03	pos.	+8.39 [+7.86, +8.91]
	Noisy class-mean bank	+5.68	pos.	+6.03 [+5.38, +6.67]

Study	c	Two-sided	Text half	Image half	Interaction [95% CI]
Follow-up A, $\sigma = 0.25$	0.25	+0.79	+0.03	+0.85	−0.09 [−0.59, +0.41]
	0.5	+1.09	−0.21	+0.82	+0.48 [−0.14, +1.11]
	1	+0.97	−0.27	+1.55	−0.30 [−1.09, +0.48]
Follow-up B, $\sigma = 0.12$	0.25	+0.12	−0.25	+0.19	+0.19 [−0.00, +0.37]
	0.5	−0.09	−0.68	+0.21	+0.38 [+0.12, +0.64]
	1	−0.18	−0.68	+0.30	+0.19 [−0.11, +0.49]
Follow-up B, $\sigma = 0.25$	0.25	−0.01	−0.52	+0.39	+0.12 [−0.06, +0.30]
	0.5	−0.30	−1.33	+0.45	+0.58 [+0.32, +0.84]
	1	−0.35	−0.93	+0.21	+0.36 [+0.09, +0.64]

TABLE XIV

STEP RESPONSE OF THE TWO PROPOSAL OBJECTIVES (Q3, REGISTERED TO BE REPORTED WHATEVER ITS SHAPE), POOLED OVER THE CELLS OF FOLLOW-UPS A AND B: THE PERCENTAGE OF CONFIRMATION DRAWS WHOSE ORIGINAL TOP-1 MARGIN IS AT MOST $2 \min\{1, \|w\|\}$, THE COARSE FLIP BUDGET OF PROPOSITION 2 (THE SAME FOR BOTH OBJECTIVES), THE PERCENTAGE WHOSE VOTE CHANGES, AND THE NET PERCENTAGE OF USEFUL (WRONG-TO-CORRECT) MINUS HARMFUL (CORRECT-TO-WRONG) CHANGES. THE COARSE BUDGET IS A NECESSARY CONDITION: IT ADMITS FAR MORE DRAWS THAN CHANGE.

Step	In flip budget		Boundary-active				Noisy-margin			
	A	B	changed A	changed B	net A	net B	changed A	changed B	net A	net B
0.0025	44.83	24.37	0.39	0.24	+0.01	+0.00	0.21	0.16	−0.01	−0.00
0.005	65.98	36.65	0.77	0.49	+0.02	+0.01	0.41	0.32	−0.02	−0.00
0.01	84.90	49.47	1.51	0.97	+0.04	+0.01	0.83	0.64	−0.04	−0.01
0.02	95.67	61.29	2.92	1.91	+0.05	+0.02	1.69	1.28	−0.09	−0.01
0.04	99.86	76.07	5.50	3.71	+0.03	+0.00	3.41	2.56	−0.17	−0.02
0.08	100.00	97.60	9.85	6.90	−0.08	−0.07	6.76	4.99	−0.26	+0.02
0.16	100.00	100.00	16.51	11.54	−0.36	−0.42	13.05	9.43	−0.43	+0.13
0.32	100.00	100.00	26.59	18.61	−0.92	−1.66	25.58	17.11	−1.04	+0.05

APPENDIX E

EXECUTION DEVIATIONS AND VERIFICATION STATUS

A. *Audit and positive control*

According to the archived review records, a second author within the team reconstructed the audit’s result chain before the post-hoc analyses: all 216 banks from the development tensors and all 59,400 outcomes from the saved counts, with data-role separation and the absence of test access checked. The standard-metric reconstruction, the simultaneous bands and the per-example counts come from released scripts, and AI-assisted reviews within the team reproduced the band output byte for byte. Other authors replayed the positive control from its raw files, as did an AI-assisted replay. The review ledger released with the artifact records, for every check, its reviewer (by role in the anonymous version), version, scope, date and AI assistance, and whether the reviewer confirmed the record in writing.

B. *Registered extension*

The extension’s registrations have digests 9802bf3e5f97 (Extension A), 3fcdade664a (Extension B) and 96fd42f268d9 (fresh-noise diagnostic). Its queue completed all 36 cells (terminal receipt PASS, 76 result files, manifest 52bfe582671c), and every cell merged and verified against its registration. The values of Section VII are primary-verified: a deterministic reconstruction from the raw shards, released with the artifact, reproduces every registered output to machine precision, and replays within the team, by other authors and AI-assisted, repeated it. Execution deviated from the registration text in three disclosed ways: the preparation stage (development features, directions and controls) left one reduced-precision convolution mode enabled, while certification ran at full precision; the registered bootstrap names one pseudo-random generator while the analysis code uses NumPy’s default generator; and the extension bound model checkpoints by name and pretrained tag rather than by file hash, so the cached checkpoint files were hashed after the run (they equal the preflight copies) but the shards do not cryptographically attest the bytes they loaded. The queue was interrupted twice, by a server reboot and by a worker crash, and resumed each time under a documented recovery without any protocol change.

C. *Prospective follow-up*

The follow-up of Section VIII ran on GPUs of a single class under the pinned environment of its data preflight, checked before every job. One GPU ran far below its normal clock and was stopped; its one started job was recomputed from its first item and its other jobs ran on GPUs of the same class, so no reported certificate comes from it, and jobs not yet started on slower GPUs were moved to idle ones. Every noise draw derives from its item’s registered seed, so neither change altered a draw, a registered quantity, an approval or an item. The registered analysis ran once, after a recorded approval that binds the run custody manifests; its results are primary-verified and were replayed within the team from the released vote counts.

The registered bootstrap compares floating-point replicates with each test boundary (zero or ± 2 points), so when an estimate lies exactly on a boundary, rounding can put tied replicates on either side. An AI-assisted recomputation within the team with exact integer numerators found this in nine raw tail probabilities outside the primary family; no direction or magnitude class and no printed value changes. The registered outputs stay as released; the exact recomputation is released with the artifact.

The registrations put the observed paired discordance next to every interval. The registered analysis did not compute it for the derived intervals of Table XIII (a supervised control against two-sided centering, and the text/image interaction), so they are reported without it; this is a reporting deviation, and for the four-arm interaction no single paired discordance is defined.